SCALE-COVARIANT ATTENTION FOR TRAINING-FREE HIGH-RESOLUTION DIFFUSION TRANSFORMERS

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ABSTRACT

Diffusion transformers sampled far above their training resolution see more tokens and longer relative positions than in training; at 4096^2, full attention in FLUX.1-dev costs about 256 times more than at 1024^2, and its images collapse into stripes. Training-free fixes restrict which keys a query sees, but give every head the same receptive field on a hand-set schedule. We measure what each head needs instead. At the training resolution, where full attention can be trusted, we emulate a larger image by shrinking a native-extent window and a subsampled view of the whole image by the same factor, and fit in closed form how much of each head's output each view explains. In FLUX.1-dev this split differs strongly between heads, is reproduced by disjoint prompt sets (split-half correlation 0.99), and transfers across emulated scales and calibration resolutions at no more than 5.4% extra output error. Scale-Covariant Attention (SCA) applies the fitted table at the target resolution: each head mixes a global branch on the native grid with an exact sliding window, and both branches keep the native key count and image offsets. An opening phase uses the global branch alone, because no window can resolve layout at high noise. SCA needs one calibration pass of about 1.5 GPU-hours and no training, and its attention cost is linear in the number of tokens. On 191 Aesthetic-4K prompts at 4096^2, under one protocol shared with AReA's released code, SCA raises ImageReward from 0.92 to 1.23 and HPSv2.1 from 27.2 to 29.7, lowers FID from 146.0 to 138.4, is significantly better on every preference, alignment, and no-reference quality metric we report, and runs 3.0 times faster than AReA and 2.6 times faster than full attention. The gain grows with the scale factor: SCA ties AReA at 2048×4096 and raises ImageReward from 0.39 to 0.66 at 2048×6144. [TODO: E1, E8: one sentence on RoPE baselines and the second model.]

1 INTRODUCTION

Text-to-image diffusion transformers such as FLUX.1 (Black Forest Labs, 2024) and Stable Diffusion 3 (Esser et al., 2024) are trained at about one megapixel. At 4096×4096 the number of image tokens grows 16-fold and the cost of full attention about 256-fold, and every attention layer sees a token count and relative positions outside its training distribution. Outputs show duplicated subjects, broken layouts, and blurred texture (He et al., 2023; Du et al., 2023; Issachar et al., 2025; Rajabi et al., 2026a); under our protocol, full attention in FLUX.1-dev at 4096^2 gives fields of stripes (Figure 1, bottom row).

Training-free fixes edit the positional encoding (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; Rajabi et al., 2026b) or restrict the keys each query sees (Zhang et al., 2023; Ma et al., 2025; Xue et al., 2026; Rajabi et al., 2026a). AReA (Rajabi et al., 2026a), the most recent of the second kind, attends to keys subsampled onto a native-size grid for the first 8 of 28 steps, then to windows that shrink to native size, with several repairs for windowing artifacts. Every head of every layer gets the same receptive field at a given step, on a schedule set by hand from attention statistics averaged over heads and layers.

We ask the model instead. At its training resolution, how much of each head's output does a native-size neighbourhood explain, and how much does a downsampled view of the whole image explain? We emulate a target s times larger by shrinking both views by s, so that the ratio of image to window matches the target while full attention stays in distribution and serves as ground truth. A convex mix

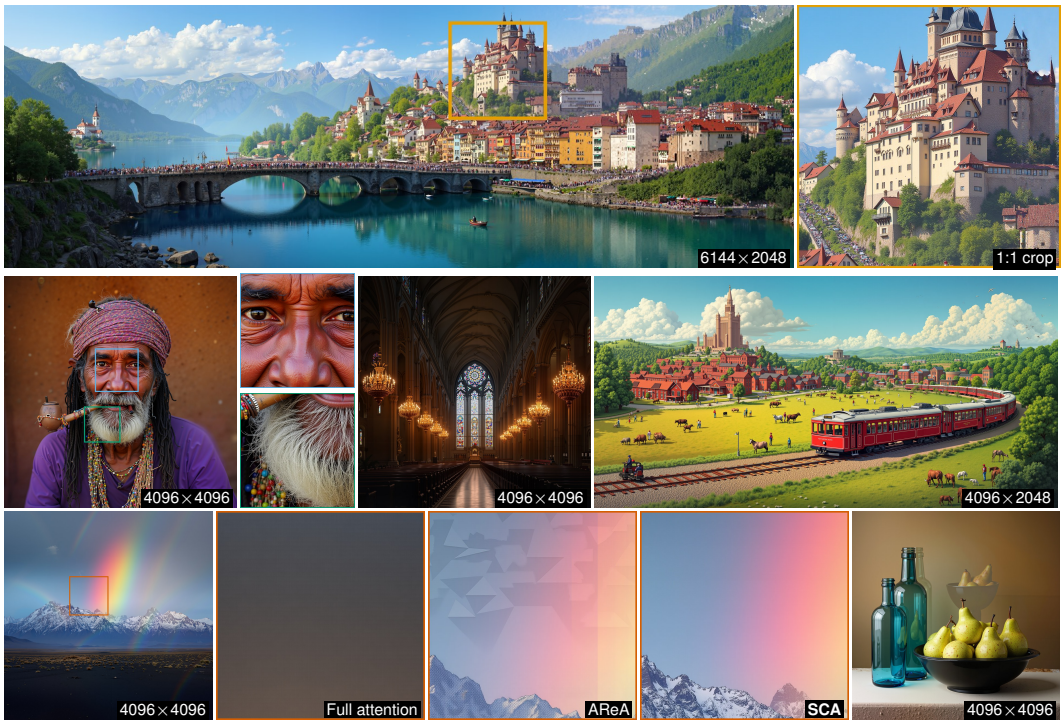

Figure 1: **SCA generates coherent, detailed images up to 6144×2048 from FLUX.1-dev without training.** All images are SCA outputs at seed 0 (labels give width×height); framed tiles are 1:1 crops at native resolution from the outlined windows. Bottom row: the same prompt and seed at 4096×4096 with full attention, which collapses into a blurred field, with AReA, whose sky breaks into tiled facets, and with SCA.

of the two views fits each head well. Its weight differs strongly between heads, is a property of the model rather than of the prompts (split-half correlation 0.99), and transfers: the table fitted at emulated scale 2 explains scale 4 with 2.4% more output error than the scale-4 table, and a 1024^2 table explains 1280^2 with 0.8% more (Section 3). We call this scale covariance: as the image grows, the local view follows a law fixed in tokens, the global view a law fixed in image fractions, and the weight between them stays put.

Scale-Covariant Attention (SCA) turns this measurement into an attention operator (Figure 2). Each head mixes a global branch, which attends to block-centre keys on the native grid with rescaled coordinates, and a local branch, an exact native-extent sliding window, with its calibrated weight. Both branches keep the native key count and every image query-key offset inside the trained range (Proposition 1), their cost is linear in the number of tokens (Proposition 2), and the calibrated weight is the ℓ_2-optimal mix at the emulated scale (Proposition 3). Output matching cannot set the opening steps: applied from the first step at 4096^2, the calibrated mix draws five Moai statues for a prompt asking for one (Figure 5), because at high noise only the lowest spatial frequencies carry signal and a 64-token window spans a quarter of the image (Argument 1). SCA therefore runs the global branch alone for the first 8 of 28 steps. Our contributions are:

1. **A calibration principle** for training-free resolution extrapolation: emulate the target scale inside the training distribution and fit each head's receptive-field mix to the model's own full attention in closed form, from statistics collected in one pass. On FLUX.1-dev the mix is head-specific, prompt-stable, and transfers across scale and calibration resolution.

2. **The SCA operator**: a per-head, per-noise-level mix of two branches that stay in the trained position range, an opening phase motivated by a frequency argument, and an exact FlexAttention sliding window; one attention layer at 4096^2 takes 18.7 ms against 87.3 ms for full attention.

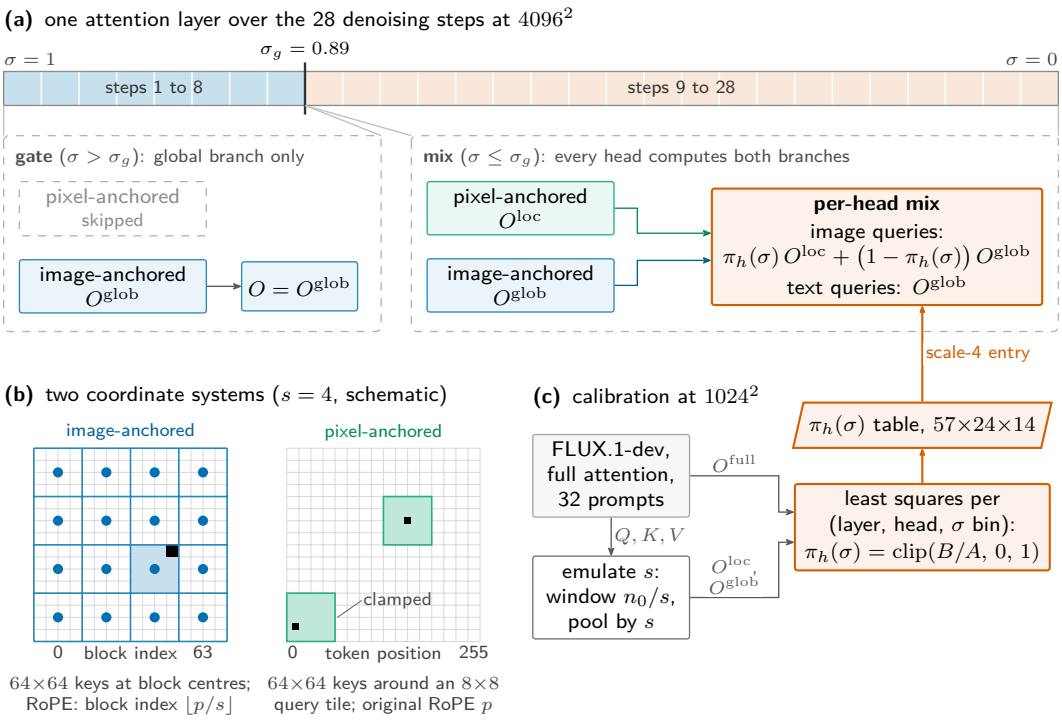

Figure 2: **SCA keeps the global branch alone for the first 8 steps, then mixes two in-distribution branches per head with a calibrated weight.** (a) One attention layer at 4096^2 over 28 denoising steps. While $\sigma > \sigma_g = 0.89$ (steps 1 to 8), every head uses only the image-anchored output O^{glob} and the pixel-anchored branch is skipped. Afterwards each head computes both branches; image queries receive $\pi_h(\sigma)\, O^{\text{loc}} + (1 - \pi_h(\sigma))\, O^{\text{glob}}$ and text queries receive O^{glob}. (b) The two coordinate systems at $s = 4$, drawn schematically. Image-anchored: keys and values are subsampled at the centre of every $s \times s$ block, giving a 64×64 grid like the native 1024^2 one, and RoPE uses block indices, so a query (black) takes the position of its block (shaded). Pixel-anchored: each 8×8 query tile attends to the 64×64 image keys around it, clamped at the border, with the original RoPE positions. Both branches always include all text keys. (c) $\pi_h(\sigma)$ is fit once per model at 1024^2: FLUX.1-dev runs with full attention on 32 prompts, scale s is emulated with a window of n_0/s and pooling by s, and closed-form least squares against the model's own O^{full} gives one weight per layer, head, and σ bin. Inference at 4096^2 reads the scale-4 entry.

3. **An evaluation under one shared protocol** with paired bootstrap intervals for every metric. On Aesthetic-4K, SCA improves every preference and quality metric over AReA's released code at 4096^2 in a third of its time, and its advantage grows with the scale factor; [TODO: E1: DyPE, SEGA, UltraImage]; [TODO: E2: 6144^2, 8192^2]; [TODO: E4: HRGen-1K]; [TODO: E8, E9: FLUX.2-dev, FLUX.1-Krea-dev].
4. **Ablations** showing that the opening phase, the per-head weights, and block-centre subsampling are each needed, and that per-layer and shared weights lose in the order their output error predicts; [TODO: E5, E6, E7: seeds, calibration budget, gate sensitivity].

2 Scale-Covariant Attention

2.1 Setting and Notation

A multimodal diffusion transformer such as FLUX.1 concatenates T text tokens $\mathcal{T}$ with $N = HW$ image tokens on the grid $\mathcal{G} = [H] \times [W]$, $[n] = \{0, \dots, n-1\}$, and applies joint attention in each block (57 blocks of 24 heads in FLUX.1). It is trained at about 1024^2 pixels, a native grid of $n_0 \times n_0$ tokens with $n_0 = 64$. Rotary embeddings give token j a position $p_j \in \mathbb{Z}^2$ (FLUX.1 puts all text tokens at 0) and a rotation R_p with $R_p^\top R_{p'} = R_{p'-p}$. For one head, a pre-rotation query q placed at

u, and a set $\mathcal{K}$ of (key, value, key position) triples,

$$\mathcal{A}(q, u; \mathcal{K}) \;=\; \textstyle\sum_{(k,v,u')\in\mathcal{K}} \operatorname{softmax}_{\mathcal{K}}\big(q^\top R_{u'-u}\, k/\sqrt{d}\big)\; v. \tag{1}$$

Full attention is $O_i^{\text{full}} = \mathcal{A}\big(q_i, p_i; \{(k_j, v_j, p_j)\}_{j\in\mathcal{T}\cup\mathcal{G}}\big)$. The target grid is $H = s_H n_0$, $W = s_W n_0$ with integer factors (s when equal), so full attention meets offsets s times the trained range. The latent at noise level σ is $x_\sigma = (1-\sigma)x_0 + \sigma\epsilon$. Softmax attention over a partition of the keys into A and B splits exactly as $O_i^{\text{full}} = m_i O_i^A + (1-m_i)O_i^B$, with m_i the softmax mass on A (Lemma 1): every head is a query-dependent mix of a near and a far field. SCA replaces both restricted outputs with in-distribution operators and m_i with a calibrated per-head weight.

2.2 Two in-distribution operators

Global operator. Let $\mathcal{G}_0 = [H/s_H] \times [W/s_W]$ be the $n_0 \times n_0$ grid of blocks. Block-centre subsampling keeps for each block b the token at $\kappa(b) = (s_H b_1 + \lfloor s_H/2 \rfloor,\, s_W b_2 + \lfloor s_W/2 \rfloor)$, and the coarse map $c(p) = (\lfloor p_1/s_H \rfloor, \lfloor p_2/s_W \rfloor)$ sends a position to its block, so $c(\kappa(b)) = b$:

$$\mathcal{K}^{\text{glob}} = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_{\kappa(b)}, v_{\kappa(b)}, b)\}_{b\in\mathcal{G}_0}, \qquad O_i^{\text{glob}} = \mathcal{A}\big(q_i,\, c(p_i);\, \mathcal{K}^{\text{glob}}\big), \tag{2}$$

with $c(p_i)$ replaced by 0 for text queries. This is AReA's first phase with the pooling factor fixed at s. It models content whose extent is a fixed fraction of the image: layout, object count, identity.

Local operator. Image queries form tiles of $t \times t$ tokens ($t = 8$); query i lies in tile $\tau = \lfloor p_i/t \rfloor$. With window side $w = \min(n_0, H)$, the window of tile τ starts at $a(\tau) = \operatorname{clip}\big(t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor, 0, H - w\big)$ per axis, centred on the tile and clamped to the grid. With $\mathcal{W}(i)$ the $w \times w$ window of i's tile,

$$\mathcal{K}_i^{\text{loc}} = \{(k_j, v_j, 0)\}_{j\in\mathcal{T}} \cup \{(k_j, v_j, p_j)\}_{p_j\in\mathcal{W}(i)}, \qquad O_i^{\text{loc}} = \mathcal{A}\big(q_i,\, p_i;\, \mathcal{K}_i^{\text{loc}}\big), \tag{3}$$

an exact sliding window with the original positions. It models content whose extent is fixed in tokens: edges, texture, fine structure.

Proposition 1 (Trained position range)**.** *Let $t \le n_0 \le \min(H, W)$. Every query-key offset of the global operator and every image query-key offset of the local operator is at most $n_0 - 1$ per axis, and each image query sees $T + n_0^2$ keys in either operator, as in full attention on the native grid.*

Proposition 2 (Cost)**.** *Per head and layer, full attention scores $(N+T)^2$ query-key pairs, the global operator $(N+T)(T+N_0)$ with $N_0 = |\mathcal{G}_0| = n_0^2$, and the local operator $N(T+w^2)$. For fixed n_0 and T the total is $O\big(N(T+w^2+N_0)\big)$, linear in N.*

Proofs are in Appendix A. Text-image offsets in the local operator equal p_i, as in full attention at the target (Remark 1). At 4096^2 SCA scores 7.2 times fewer pairs than full attention, 8.4 times fewer with the opening phase (Table 6).

2.3 Per-head mixing and the opening phase

For an image query of head h in block l at noise level σ,

$$O_i \;=\; O_i^{\text{glob}} \;\text{ if } \sigma > \sigma_g, \qquad O_i \;=\; \pi_{lh}(\sigma)\, O_i^{\text{loc}} + \big(1 - \pi_{lh}(\sigma)\big)\, O_i^{\text{glob}} \;\text{ otherwise}, \tag{4}$$

and text queries use O_i^{glob}. The weight $\pi_{lh}(\sigma)$, written $\pi_h(\sigma)$, stands in for the near-field mass m_i; one weight per head is cheap and a few prompts determine it.

Argument 1 (Why output matching cannot set the opening steps). At noise level σ a spatial frequency band has signal-to-noise ratio $\big((1-\sigma)/\sigma\big)^2 S(f)$ for image spectrum S. Natural spectra fall with frequency, so early on only the lowest band carries signal, and it fixes layout and object count. A window of w tokens holds less than one period of any frequency below $1/w$, so it cannot tell one large object from a repeating pattern; neighbouring windows settle the question independently and duplicate subjects at the window's scale. The global operator sees the whole image and resolves this band. Squared output error is dominated by texture, so calibration cannot detect the problem.

We have not measured the per-band signal-to-noise ratio of FLUX.1 latents, but the argument's prediction holds: a 16-token window gives smaller and more numerous copies than a 64-token one (Figure 5). We set $\sigma_g = 0.89$, the first 8 of 28 steps at maximum shift, which is also AReA's switch point; gating 13 steps gives a 4×4 block mosaic, because all queries of a block share one coarse position.

2.4 CALIBRATION BY OUTPUT MATCHING

The weights must come from where the model is trustworthy, which the target is not. On an in-distribution $n \times n$ grid ($n = 64$ at 1024^2), emulated scale s_c uses subsampling factor s_c, window side n/s_c, and tile side $\mathrm{round}(tn/(s_c n_0))$, so that the ratio of grid to window is s_c, as at a target of scale s_c. The sampler runs unmodified full attention; at every block and step both emulated operators are evaluated on the same queries, keys, and values. For block l, head h, and noise bin β, sums over prompts, steps with $\sigma \in \beta$, image queries, and channels give

$$A = \sum \|O^{\mathrm{loc}} - O^{\mathrm{glob}}\|^2, \quad B = \sum \langle O^{\mathrm{full}} - O^{\mathrm{glob}}, O^{\mathrm{loc}} - O^{\mathrm{glob}} \rangle, \quad E = \sum \|O^{\mathrm{full}} - O^{\mathrm{glob}}\|^2, \tag{5}$$

and the squared error of the mix with weight π is $J(\pi) = E - 2\pi B + \pi^2 A$.

Proposition 3 (Closed-form calibration and transfer cost)**.** *In each cell (l, h, β), $\pi^\star = \mathrm{clip}(B/A, 0, 1)$ ($\pi^\star = 1$ if $A = 0$) minimizes J on $[0, 1]$. Any other weight $\pi' \in [0, 1]$ (fitted at another scale or resolution, per layer, shared, or a hard switch) has $J(\pi') - J(\pi^\star) \geq A(\pi' - \pi^\star)^2$, with equality when $0 < \pi^\star < 1$.*

So the per-head table has the lowest output error of any rule constant within a cell, and replacing it costs an A-weighted squared distance. This holds at the emulation scale; Section 3 measures empirically how well tables transfer between scales and resolutions. The same sums score every alternative rule without another forward pass. We use 14 bins of width $1/14$ in σ and interpolate linearly between bin centres at inference. One pass collects emulated scales 2 and 4; calibration took about 1.5 GPU-hours on one GH200 for 32 prompts at 1024^2 and 16 at 1280^2. A target with factors (s_H, s_W) uses the table whose scale is nearest $\max(s_H, s_W)$ in log space, here always 4. The local operator is one FlexAttention call (Dong et al., 2024) over tile-major tokens with the window membership as block mask, so the window is exact. Algorithm 1 and implementation details are in Appendix C.

3 WHAT THE HEADS WANT

We calibrate FLUX.1-dev on 32 prompts at 1024^2 and 16 at 1280^2, written for this purpose and disjoint from the evaluation sets. Errors are relative, $\sum J / \sum \|O^{\mathrm{full}}\|^2$, pooled over blocks, heads, and steps.

Heads differ. Over the 1368 heads, the standard deviation of $\pi_h(\sigma)$ is 0.24 to 0.26 at every noise level (emulated scale 2); half of all (head, noise level) cells lie in $[0.2, 0.8]$ (50.4%; 49.7% at scale 4), 11.7% exceed 0.95, and 2.9% are below 0.05 (Figure 3). The mean weight is 0.64 to 0.74 (0.68 to 0.77 at scale 4): the typical head leans local but still draws on the global branch.

The table is a property of the model. Tables fitted on disjoint halves of the prompts correlate at 0.989 over all cells (0.986 at scale 4; 0.974 for 16 prompts at 1280^2), and each half's table scored on the other half gives a relative error of 0.0576 against 0.0574 in sample. [TODO: E6: tables from 4, 8, 16 prompts.]

The weights barely depend on scale or calibration resolution. The scale-2 table applied at scale 4 gives a relative error of 0.205 against 0.200 for the scale-4 table, still below every shared rule fitted at scale 4. The 1024^2 table scored at 1280^2 gives 0.0589 against 0.0585. Over all pairs of calibration targets, a transferred table adds at most 5.4% error, while each target's own per-layer fit adds 12 to 24%. Since the two resolutions emulate scale 2 with windows of 32 and 40 tokens, the weights follow the ratio of image to window, not the window's size.

Per-head beats coarser rules; output error misses layout. Relative to the best hard switch, per-head mixing lowers the relative error from 0.092 to 0.057 at scale 2 and from 0.261 to 0.200 at scale 4; per-layer tables reach 0.071 and 0.223, one shared weight per noise level 0.077 and 0.230 (Figure 7). In output energy the best hard switch never switches (Appendix B); the opening phase overrides this on purpose (Argument 1).

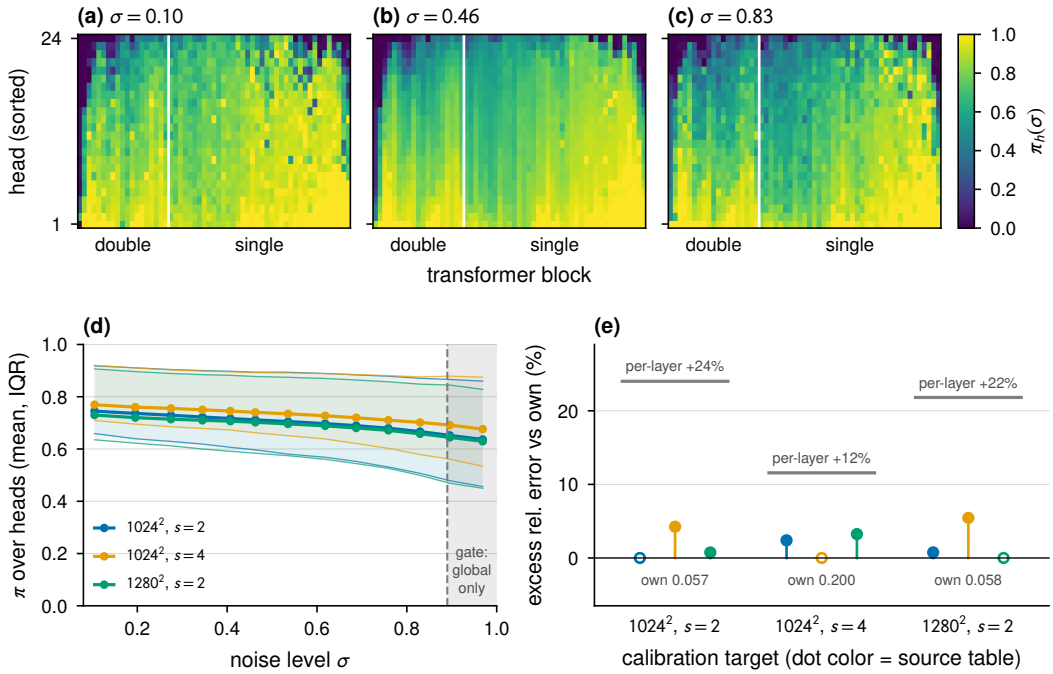

Figure 3: **Mixing weights vary strongly across heads, yet a calibrated table transfers across scale and resolution at little cost.** (a to c) Local weight $\pi_h(\sigma)$ of the scale-4 table used at inference, for all 57 blocks and 24 heads at three noise levels below the gate ($\sigma = 0.89$). Heads are sorted within each block by their mean weight, in the same order in every panel; the white line separates the 19 double-stream blocks from the 38 single-stream blocks. Below the gate, 48.7% of the cells lie in $[0.2, 0.8]$ and the standard deviation over heads is 0.22. (d) Mean of π over all heads, with thin lines and light fill marking the interquartile range, for three calibrations: 1024^2 with 32 prompts at emulated scales 2 and 4, and 1280^2 with 16 prompts at scale 2. In the shaded region the gate replaces the table with the global branch. (e) Relative output error of each per-head table on each calibration target, as excess over the target's own table (hollow dot; its absolute error is printed below). A transferred table adds at most 5.4% error, while the target's own per-layer fit (grey bar) adds 12% to 24%.

4 Experiments

Protocol. The sampler uses AReA's FLUX.1 defaults for every method, and AReA runs from its released code with FLUX.1-dev (Appendix I). SCA uses $t = 8$, $w = 64$, $\sigma_g = 0.89$, and the scale-4 table from 32 prompts at 1024^2. "SCA + band guard" was our first full configuration (Appendix H).

4.1 Main results

At 4096^2, SCA beats AReA on every preference, alignment, and no-reference quality metric (Table 2): ImageReward $+0.31$ $[0.24, 0.38]$, HPSv2.1 $+2.50$ $[2.17, 2.88]$, PickScore $+0.73$ $[0.62, 0.86]$, CLIP score $+1.45$ $[1.06, 1.82]$, MUSIQ$_c$ $+10.9$ $[9.5, 12.4]$. FID falls by 7.6 $[3.5, 12.5]$ and patch KID falls significantly; patch FID, KID, and Inception score are within noise. SCA takes 71.0 s per image, 2.6 times less than full attention and 3.0 times less than AReA. Full attention collapses into stripes and is significantly worse than AReA on every metric. At 1:1 scale, AReA's skin and fur are smoother than SCA's and its surfaces carry speckle and tiling artifacts (Figure 4). [TODO: E1: compare with DyPE, SEGA, UltraImage on quality and latency.]

Other aspect ratios. The advantage grows with the scale factor. At 2048×4096 (factors 2 and 4) SCA ties AReA: ImageReward (-0.00 $[-0.08, 0.07]$) and FID are within noise, and SCA is significantly better on three metrics and worse on three. At 2048×6144 (factors 2 and 6) SCA

Table 1: **Evaluation protocol.** All methods share sampler, seeds, and decoder, so only attention differs. Prompt counts exclude four development prompts. Appendices D and E give hyperparameters and metric definitions.

Models	FLUX.1-dev (Black Forest Labs, 2024) (main results); FLUX.1-Krea-dev (Black Forest Labs & Krea AI, 2025) [TODO: E9]; FLUX.2-dev (Black Forest Labs, 2025) [TODO: E8]; bf16.
Sampler	28 Euler flow-matching steps, guidance 4.5, shift $\mu = 1.15$ at every size, seed 0 per prompt for all methods, tiled VAE decoding.
Sizes	4096^2, 2048×4096, 2048×6144 (191 prompts each); 6144^2 (100 prompts) and 8192^2 (qualitative) [TODO: E2].
Prompts	Aesthetic-4K evaluation set with 4K references (Zhang et al., 2025a), 191 prompts, fixed 46-prompt ablation subset; HRGen-1K (Rajabi et al., 2026a), 200 prompts [TODO: E4].
Baselines	Full attention; AReA, released code (Rajabi et al., 2026a); DyPE (Issachar et al., 2025), SEGA (Rajabi et al., 2026b), UltraImage (Zhao et al., 2025) [TODO: E1].
Metrics	ImageReward, HPSv2/2.1, PickScore, CLIP, aesthetic; FID, KID, IS, patch FID/KID; MUSIQ (full, 1024^2 crops), CLIP-IQA, ARNIQA; 95% paired bootstrap intervals over prompts.
Hardware	One NVIDIA GH200 (96 GB) per process; latency at batch 1 on an idle node.

Table 2: **Aesthetic-4K with FLUX.1-dev**, 191 prompts (100 at 6144^2). HPS, CLIP-IQA, and ARNIQA are $\times 100$; higher is better except latency and FID. Bold: best per size. * / †: significantly better / worse than AReA (95% paired bootstrap interval excludes zero). Latency at 4096^2: seconds per image, batch 1, idle GH200, median of three. ‡Mean campaign time on shared nodes. **?**: pending. Appendix F gives all values with intervals.

Method	Lat. (s)↓	IR	Aesth.	CLIP	PickS.	HPSv2	HPSv2.1	FID↓	FID_p ↓	MUSIQ	MUSIQ_c	CLIP-IQA	ARNIQA
4096×4096, scale factors (4, 4)													
Full attention	184.6	$-2.18^\dagger$	$4.51^\dagger$	$15.38^\dagger$	$17.94^\dagger$	$20.75^\dagger$	$7.61^\dagger$	$312.7^\dagger$	$257.9^\dagger$	$22.9^\dagger$	$33.2^\dagger$	$49.0^\dagger$	$54.5^\dagger$
DyPE [TODO: E1]	?	?	?	?	?	?	?	?	?	?	?	?	?
SEGA [TODO: E1]	?	?	?	?	?	?	?	?	?	?	?	?	?
UltraImage [TODO: E1]	?	?	?	?	?	?	?	?	?	?	?	?	?
AReA (released code)	212.6	0.92	6.28	27.67	22.35	28.64	27.20	146.0	110.2	39.4	50.4	67.5	67.3
SCA + band guard	88.1	1.20^*	**6.46***	28.87^*	22.90^*	29.40^*	28.95^*	142.3	**104.8***	41.4^*	61.0^*	70.9^*	70.8^*
SCA (final)	**71.0**	**1.23***	6.45^*	**29.11***	**23.09***	**29.77***	**29.71***	**138.4***	106.5	**45.3***	**61.2***	**72.9***	**71.2***
2048×4096, scale factors (2, 4)													
AReA (released code)	$112.4^\ddagger$	**1.13**	**6.41**	**28.51**	**22.67**	28.38	**27.66**	144.8	**119.0**	51.8	61.6	**74.8**	**73.0**
SCA (final)	**38.1**‡	**1.13**	$6.21^\dagger$	28.21	22.64	**28.55***	27.46	**143.8**	$124.1^\dagger$	**55.9***	**65.9***	$73.5^\dagger$	72.8
2048×6144, scale factors (2, 6)													
AReA (released code)	$150.5^\ddagger$	0.39	**6.15**	26.49	21.51	26.41	23.35	163.6	**112.8**	42.9	53.1	67.3	69.3
SCA (final)	**55.8**‡	**0.66***	$6.00^\dagger$	**27.01**	**21.74***	**26.54***	**23.48**	**143.8***	116.5	**48.4***	**60.6***	**69.0***	**70.2***
6144×6144, scale factors (6, 6), 100 prompts [TODO: E2]													
AReA (released code)	?	?	?	?	?	?	?	?	?	?	?	?	?
SCA (final)	?	?	?	?	?	?	?	?	?	?	?	?	?

is significantly better on ImageReward ($+0.27$ $[0.15, 0.40]$), FID (-19.8 $[-24.5, -11.9]$), KID, Inception score, PickScore, HPSv2, both MUSIQ variants, CLIP-IQA, and ARNIQA, and worse only on aesthetic score (-0.15). The FID gap grows from -1.1 to -7.6 to -19.8 across the three sizes. The scale-2 table changes the 2048×4096 results negligibly (Appendix G). [TODO: E2: 6144^2 metrics and 8192^2 images.]

Latency. SCA's attention is linear in N (Proposition 2). At 4096^2 one attention layer takes 18.7 ms against 87.3 ms for full attention, and a whole image 71.0 s against 184.6 s (Table 3). [TODO: E3: the curve from 2048^2 to 8192^2.]

4.2 Ablations

Table 5 changes one component at a time (examples in Figure 14). **The opening phase is essential:** without it (v0), ImageReward falls by 1.33 $[1.06, 1.60]$ and subjects repeat; CLIP-IQA and ARNIQA rise, since each copy is sharp. **The per-head mix carries global context after the opening phase:** local-only attention in the last 20 steps lowers ImageReward by 0.36 $[0.21, 0.52]$, HPSv2.1 by 2.86, PickScore by 1.20, and CLIP score by 1.93. **Per-head weights beat shared and per-layer weights,**

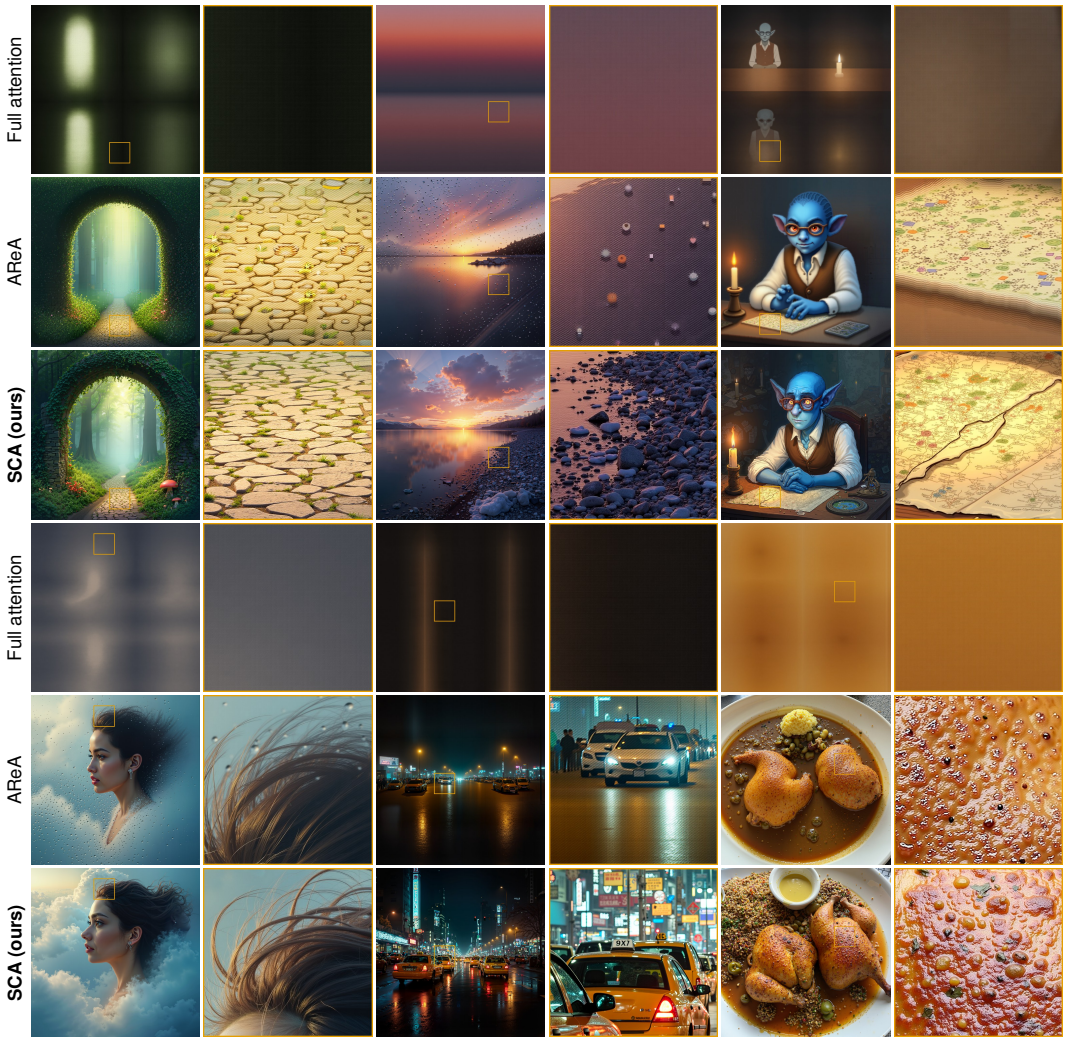

Figure 4: **At 4096×4096, full attention collapses, AReA loses detail and adds texture artifacts, and SCA keeps both layout and fine detail.** FLUX.1-dev, same prompt and seed (0) per column pair; each full image is followed by a 1:1 crop of its outlined 512×512 window. Crop windows are chosen automatically as the region with the most edge energy averaged over the three methods. Prompts were picked for variety from the 191 evaluation prompts; uncurated random samples are in Figure 10.

Table 3: **Seconds per image**, batch 1, idle GH200, 28 steps with text encoding and VAE decoding, median of three. [TODO: E3]

Size (N tokens)	Full	AReA	SCA
2048^2 (16,384)	?	?	?
3072^2 (36,864)	?	?	?
4096^2 (65,536)	184.6	212.6	**71.0**
6144^2 (147,456)	?	?	?
8192^2 (262,144)	?	?	?

Table 4: **Generality and sensitivity** at 4096^2: paired difference, method minus reference; intervals in Appendix F. **?**: pending.

Setting (vs. reference)	ΔIR	ΔHPSv2.1	ΔMUSIQ$_c$
E4 HRGen-1K, vs. AReA	?	?	?
E5 seeds 1, 2, vs. AReA	?	?	?
E6 4 / 8 / 16 calib. prompts, vs. 32	?	?	?
E7 $\sigma_g = 0.85$ / 0.93, vs. 0.89	?	?	?
E8 FLUX.2-dev, vs. full attention	?	?	?
E9 FLUX.1-Krea-dev, vs. AReA	?	?	?

in the order output error predicts: one shared weight per noise level (mix error 0.230) lowers ImageReward by 0.09 $[0.01, 0.17]$ and MUSIQ$_c$ by 10.4; per-layer weights (0.223) still lose significantly on HPSv2.1, PickScore, ARNIQA, and MUSIQ$_c$. **Neither branch works alone, and**

Table 5: **Ablations at** 4096^2, 46 prompts. First row: absolute values for SCA final; other rows change one thing and give the paired difference. **Bold blue** / *orange italic*: significantly better / worse; patch FID has no interval. HPS, CLIP-IQA, ARNIQA $\times 100$. "Mix err.": relative output error of the weight rule at emulated scale 4. Intervals: Table 10.

Variant	Mix err.	IR	Aesth.	HPSv2	HPSv2.1	PickS.	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ_e	$\text{FID}_p \downarrow$
SCA final (reference)	0.200	1.12	6.45	29.71	29.98	22.96	29.41	71.3	70.5	46.1	62.5	170.2
w/o per-head mixing ($\pi = 1$)	0.261	-0.36	-0.21	-1.78	-2.86	-1.20	-1.93	-4.6	-1.0	-3.5	-0.3	$+24.1$
one shared $\pi(\sigma)$	0.230	-0.09	-0.09	-0.96	-1.83	-0.38	-0.38	-7.7	-6.5	-4.9	-10.4	-1.1
per-layer $\pi_\ell(\sigma)$	0.223	-0.05	$+0.01$	-0.50	-0.70	-0.16	-0.13	-0.9	-1.5	$+0.7$	-2.4	0.0
global only ($\pi = 0$)	0.627	-0.40	-0.76	-3.32	-5.66	-1.44	-3.48	-27.0	-45.1	-3.5	-11.7	$+68.7$
mean pooling	n/a	-1.52	-0.61	-4.04	-9.75	-2.60	-3.59	-10.1	-4.8	-1.8	-4.2	$+17.3$
v0: no opening phase	0.200	-1.33	-0.08	-1.89	-4.07	-2.18	-1.65	$+2.9$	$+4.3$	-5.6	$+2.5$	$+34.7$

block centres beat block means: global only loses on every metric, and mean pooling blurs the global keys (ImageReward -1.52). Band-guard ablations, including hard per-head routing, are in Appendix H.

4.3 Generality and sensitivity

Table 4 collects the remaining experiments. [TODO: E4: HRGen-1K.] [TODO: E5: do seeds 1, 2 match seed 0?] [TODO: E6: calibration budget.] [TODO: E7: plateau or optimum at 0.89?] [TODO: E8: FLUX.2-dev and its π statistics.] [TODO: E9: FLUX.1-Krea-dev.]

5 Related work

Training-free high resolution. Earlier methods change convolutions, spectra, or attention temperature (He et al., 2023; Zhang et al., 2023; Huang et al., 2024; Jin et al., 2023), fuse global and patch paths (Du et al., 2023; Yang et al., 2024; Qiu et al., 2024), or guide with a native pass (Du et al., 2024; Bu et al., 2025; Koh et al., 2025). RoPE methods for diffusion transformers (Zhuo et al., 2024; Issachar et al., 2025; Zhao et al., 2025; 2026; Rajabi et al., 2026b; Liu et al., 2026) rescale positions but keep full attention and its quadratic cost; SCA keeps the trained positions and changes the key set. ResDiT fuses a global branch with patch attention by fixed weights (Ma et al., 2025), HRDiT gives each head a static local scope from an integer program under a cost budget (Xue et al., 2026), AReA schedules one receptive field for all heads (Rajabi et al., 2026a), and FreeScale fuses attention outputs by frequency band (Qiu et al., 2024); SCA's weights are per head and per noise level and come from the model's own full attention. **Efficient attention.** ToDo attends to downsampled keys without rescaling coordinates (Smith et al., 2024); DiTFastAttn, Sparse VideoGen, Sliding Tile Attention, and Sparse-vDiT pick windows or sparsity per layer or head at trained resolutions (Yuan et al., 2024; Xi et al., 2025; Zhang et al., 2025b; Chen et al., 2025); our local branch is neighbourhood attention (Hassani et al., 2022). **Per-head scope elsewhere.** Self-Extend groups distant positions (Jin et al., 2024), MoA assigns per-head spans (Fu et al., 2024), and DuoAttention separates retrieval heads from local ones (Xiao et al., 2024); HiLo (Pan et al., 2022) and Shunted attention (Ren et al., 2021) train vision transformers whose heads split fine and coarse keys by design. A full-attention diffusion transformer appears to learn such a split implicitly.

6 Limitations and conclusion

SCA only ties AReA at 2048×4096. Small extra figures can still appear late in sampling, and one development prompt keeps doubled limbs in every variant (Appendix K). The opening phase is set from generations, since output error cannot set it. Transfer across emulated scales supports, but cannot prove, that the weights depend only on the ratio of image to window. Main results use one model and one seed per prompt [TODO: E5, E8, E9].

Each head's split between a neighbourhood fixed in tokens and a view of the whole image can be measured once at the training resolution and reused at 4K; with it and an opening phase, SCA beats AReA's released code at 4096^2 in a third of its time.

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A PROOFS

Lemma 1 (Softmax split)**.** *Let the keys of a query be partitioned into A and B, with logits ℓ_j. Then $O^{\text{full}} = m\,O^A + (1-m)\,O^B$ with $m = \sum_{j\in A} e^{\ell_j} / \sum_j e^{\ell_j}$ and O^A, O^B the softmax outputs restricted to A and B.*

Proof. Write $Z_A = \sum_{j\in A} e^{\ell_j}$ and Z_B likewise. Then $O^{\text{full}} = (\sum_A e^{\ell_j} v_j + \sum_B e^{\ell_j} v_j)/(Z_A + Z_B) = \frac{Z_A}{Z_A+Z_B} O^A + \frac{Z_B}{Z_A+Z_B} O^B$. □

Proof of Proposition 1. Global operator. Image queries sit at $c(p_i) \in \mathcal{G}_0 = [n_0]^2$ and text queries at $0 \in \mathcal{G}_0$; subsampled keys sit at $b \in \mathcal{G}_0$ and text keys at 0. Any two points of $[n_0]^2$ differ by at most $n_0 - 1$ per axis. The key count is $T + |\mathcal{G}_0| = T + n_0^2$.

Local operator. Fix one axis of length $H \geq w$ and a tile $[t\tau,\, t\tau + t)$ with $t \leq w$. The unclamped start $a' = t\tau + \lfloor t/2 \rfloor - \lfloor w/2 \rfloor$ satisfies $a' \leq t\tau$ because $\lfloor t/2 \rfloor \leq \lfloor w/2 \rfloor$, and $a' + w \geq t\tau + t$ because $w - \lfloor w/2 \rfloor = \lceil w/2 \rceil \geq \lceil t/2 \rceil = t - \lfloor t/2 \rfloor$. If $a' < 0$ the window is $[0, w)$, and $w \geq a' + w \geq t\tau + t$. If $a' > H - w$ the window is $[H - w, H)$, with $H - w < a' \leq t\tau$ and $H \geq t\tau + t$. In every case the window contains the tile, so it contains each query of the tile, and two positions in a window of side $w \leq n_0$ differ by at most $n_0 - 1$. The key count is $T + w^2 = T + n_0^2$ when $n_0 \leq \min(H, W)$. □

Remark 1 (Text-image offsets). FLUX.1 places all text tokens at position 0, so the offset between an image query at p_i and a text key is p_i. In the global operator the query sits at $c(p_i) \in [n_0]^2$, inside the trained range. In the local operator it keeps p_i, up to $sn_0 - 1$ per axis, exactly as in full attention and AReA's windows at the target. Mapping text offsets through $c(\cdot)$ in the local operator is a possible refinement that we have not tested.

Proof of Proposition 2. Full attention scores every query against every key, $(N + T)^2$ pairs. The global operator scores $N + T$ queries against $T + |\mathcal{G}_0|$ keys with $|\mathcal{G}_0| = N/(s_H s_W) = n_0^2$. The local operator scores N image queries against T text keys and w^2 window keys. The sum is at most $(2N + T)(T + w^2 + N_0)$, which is $O(N(T + w^2 + N_0))$ for fixed n_0, T. Value aggregation has the same pair counts. In the opening phase the local term is absent. The FlexAttention kernel visits whole blocks and masks the extra pairs: a query block of 8×16 tokens spans two tiles whose windows are 8 columns apart, so it visits a 64×72 key region, 12.5% more than one window, plus partial blocks at clamped borders. This changes the constant, not the order. □

Proof of Proposition 3. $J(\pi) = E - 2\pi B + \pi^2 A$ with $A \geq 0$. If $A = 0$ then $O^{\mathrm{loc}} = O^{\mathrm{glob}}$ on every summand, so $B = 0$ and J is constant; any π, in particular $\pi^\star = 1$, is optimal and the bound holds trivially. If $A > 0$, J is a strictly convex quadratic with unconstrained minimizer B/A, and its minimizer on $[0, 1]$ is the projection $\mathrm{clip}(B/A, 0, 1)$. For any π',

$$J(\pi') - J(\pi^\star) = A(\pi' - \pi^\star)^2 + 2(\pi' - \pi^\star)(A\pi^\star - B).$$

If $0 < \pi^\star < 1$ then $A\pi^\star = B$ and the second term vanishes. If $\pi^\star = 0$ then $B \leq 0$, so $A\pi^\star - B \geq 0$ and $\pi' - \pi^\star \geq 0$. If $\pi^\star = 1$ then $B \geq A$, so $A\pi^\star - B \leq 0$ and $\pi' - \pi^\star \leq 0$. In both clipped cases the second term is nonnegative.

Any rule that assigns one weight per cell, such as a per-layer table, a weight shared per noise level, a hard switch, or a table fitted elsewhere, is a feasible π' in every cell, so its total error is at least the per-head table's, by $\sum_{\text{cells}} A(\pi' - \pi^\star)^2$ or more. A table interpolated in σ between bin centres is not constant within a bin; the statement applies to the bin-constant table used to score rules in Section 3. □

The frequency argument in more detail. Argument 1 rests on two facts and one assumption. The first fact is that the noisy latent is a sum of signal and independent white noise, so a band's signal-to-noise ratio scales as $((1 - \sigma)/\sigma)^2 S(f)$; at $\sigma = 0.89$ the factor is 0.015, and at $\sigma = 0.5$ it is 1. The second is that a window of w tokens cannot distinguish a component of period longer than w from a smooth trend within the window, so windows cannot coordinate a layout at that scale except through what earlier layers mixed into their keys. The assumption is that natural-image latent spectra fall with frequency, which we have not measured for FLUX.1 latents. Together they predict that windowed attention fails first and worst at high noise and for objects larger than the window, which matches the duplication we observe (Figure 5) and the effect of a 16-token window.

B Calibration details and statistics

Statistics. For every emulated scale, block, head, and noise bin, the pass stores A, B, E, the error of the local branch alone $\sum \|O^{\mathrm{full}} - O^{\mathrm{loc}}\|^2$, $\sum \|O^{\mathrm{full}}\|^2$, and the query count, in float64, summed over image queries and channels (`sca/calib.py`). The sampler's trajectory is that of unmodified full attention; the emulated operators only observe. Statistics are written per prompt and merged by summation, so prompts can be added later and split-half fits need no extra pass. Text-query outputs are not in the objective, because SCA sends text queries through the global branch only.

Geometry at emulated scale. At 1024^2 the grid is 64×64. Emulated scale 2 uses a 32×32 window, 2×2 subsampling onto a 32×32 grid, and 4×4 query tiles; emulated scale 4 uses a 16×16 window, 4×4 subsampling onto a 16×16 grid, and 2×2 tiles. At 1280^2 the grid is 80×80, and emulated scale 2 uses a 40×40 window, 2×2 subsampling onto a 40×40 grid, and 5×5 tiles.

Noise bins. With 28 steps and $\mu = 1.15$, σ runs from 1.0 to 0.105, so the lowest of the 14 bins is empty and copies its neighbour; 13 bins are populated. Under this schedule 8 steps have $\sigma > 0.89$ (the ninth has $\sigma = 0.888$) and 13 have $\sigma > 0.80$.

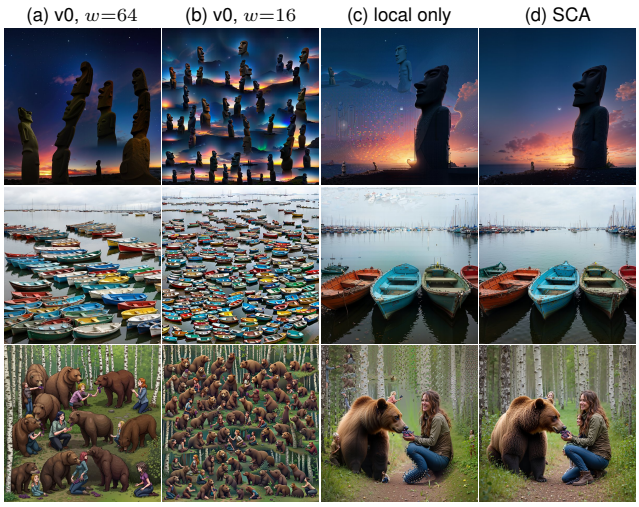

Figure 5: **Windows at high noise duplicate subjects.** 4096^2, three development prompts. (a) v0, the calibrated mix from the first step: each 64-token window settles the layout on its own. (b) A 16-token window gives smaller, more numerous copies. (c) With the opening phase but local-only attention afterwards ($\pi = 1$), ghost copies and grid speckle remain. (d) SCA.

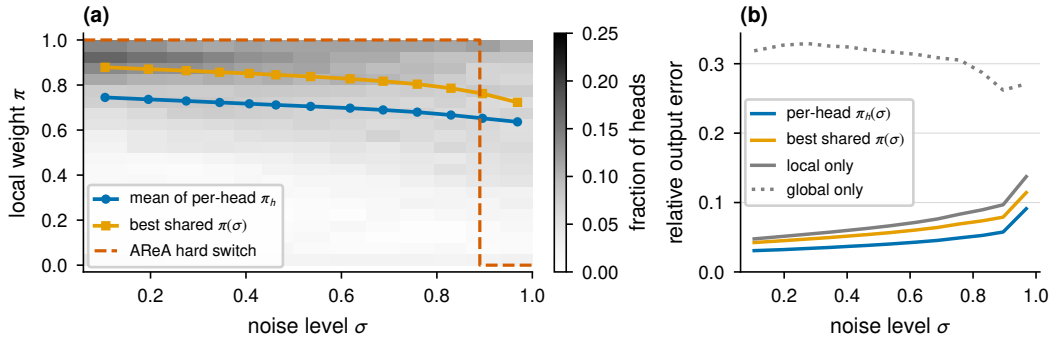

Figure 6: **In output energy, no noise level prefers the global branch.** (a) Distribution of $\pi_h(\sigma)$ over the 1368 heads per noise bin (grey), the mean over heads, and the best single shared weight, at emulated scale 2. The dashed step is AReA's schedule read as a weight. (b) Relative output error per bin. The local branch alone beats the global branch alone in every bin, and per-head mixing beats both. 32 prompts.

No global phase in output energy. The best single shared weight falls from 0.88 at $\sigma = 0.10$ to 0.72 at $\sigma = 0.97$, and the mean per-head weight from 0.74 to 0.64 (Figure 6). Among hard switches from global to local, the best threshold is $\sigma^\star = 1$, which equals local only. A switch at $\sigma^\star = 0.89$ between our two branches raises the error from 0.092 to 0.152 at scale 2 and from 0.261 to 0.376 at scale 4. In the noisiest bin, the global branch alone has relative error 0.27 at scale 2, the local branch alone 0.14, and the per-head mix 0.09. The generation results of Section 4.2 show the opposite preference for the opening steps, which is the gap between squared output error and layout.

Attention mass. For 256 random image queries per block and step, the pass also records the softmax mass on text keys and on image keys within Chebyshev distance 2, 4, 8, and 16. At 1024^2, averaged over heads and queries, text receives 15 to 19% of the mass; image keys within distance 16 receive 52% at $\sigma = 0.10$ and 40% at $\sigma = 0.97$, and keys outside that 33×33 neighbourhood still receive 29% and 41%. A purely local receptive field cannot account for any step even at native resolution.

Calibration budget. [TODO: E6: relative error of tables merged from 4, 8, and 16 prompts, scored on the held-out prompts, next to the 32-prompt table.]

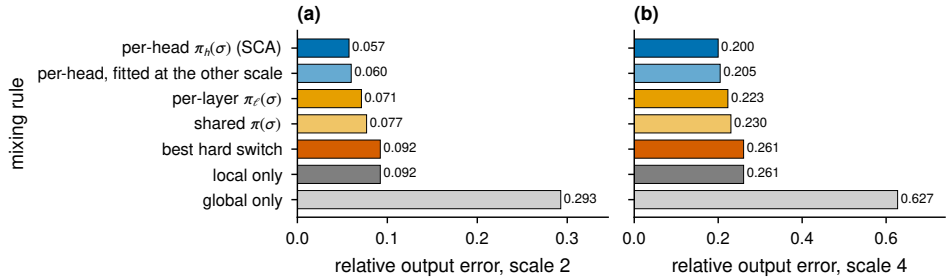

Figure 7: **Per-head weights explain full attention best, and transfer across scales.** Relative output error for each mixing rule at emulated scales 2 (a) and 4 (b), 32 prompts. "Fitted at the other scale" applies the scale-4 table at scale 2 and vice versa. The best hard switch puts its threshold at $\sigma^\star = 1$ and so equals local only. In-sample and held-out errors agree within 0.0003.

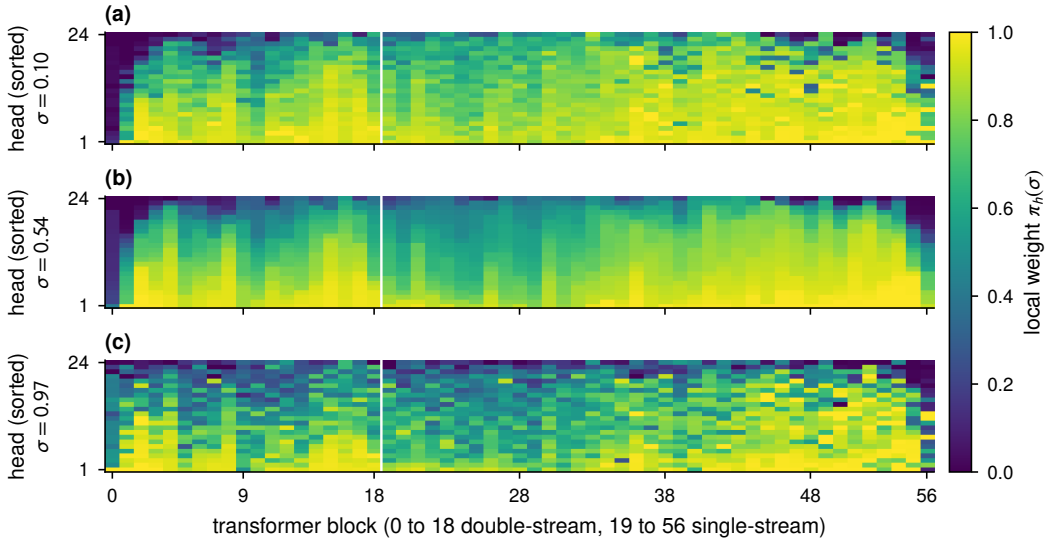

Figure 8: **Fitted weights at emulated scale 2.** $\pi_h(\sigma)$ for all 57 blocks and 24 heads of FLUX.1-dev at three noise levels, 32 prompts. Within each block, heads are sorted by their weight at $\sigma = 0.54$, in the same order in all panels. The white line separates double-stream from single-stream blocks. Figure 3 shows the scale-4 table used at inference.

C Implementation details and cost

FlexAttention window. Image tokens are reordered tile-major, so that a sparse block is a small spatial tile. Query blocks cover 8×16 tokens (128 queries) and key blocks 8×8 (64 keys), shifted so that unclamped windows start on a block edge; text keys come first. The mask function states window membership per query-key pair, so the result does not depend on the block layout, only the amount of skipped work does. On GH200, 8×16 query blocks with 8×8 key blocks were fastest at 2048^2, 4096^2, and 2048×6144. RoPE, subsampling, both branches, and the mix compile into one graph per layer; the mix is in fp32.

Exactness test. `tests/test_sca.py` (18 CPU tests) checks that the processor in full mode matches the stock FLUX attention; that both branches match a brute-force computation of Equations 2 and 3; that the local branch with a full-grid window and the global branch with factor 1 both equal full attention; that calibration returns $\pi = 1$ when the local branch is exact; that the gate returns the global branch alone; and that the FlexAttention local branch equals a reference that gathers each tile's key window explicitly, to within one bf16 unit in the last place, with both equally far from an fp32 reference.

Algorithm 1 SCA: calibration (once per model) and one attention layer at sampling time.

Calibrate(prompts, s_c) on an in-distribution $n \times n$ grid:
sample with full attention; at every block l, step σ, and head h:
  $O^{\text{glob}}, O^{\text{loc}} \leftarrow$ Eqs. 2, 3 with subsampling s_c, window n/s_c, tile $tn/(s_c n_0)$
  add $\|O^{\text{loc}} - O^{\text{glob}}\|^2$ and $\langle O^{\text{full}} - O^{\text{glob}}, O^{\text{loc}} - O^{\text{glob}} \rangle$ to $A_{lh\beta(\sigma)}$, $B_{lh\beta(\sigma)}$ ▷ float64
return $\pi_{lh\beta} \leftarrow \text{clip}(B_{lh\beta}/A_{lh\beta}, 0, 1)$ ▷ Proposition 3
Layer(Q, K, V, σ) at target factors (s_H, s_W):
$O^{\text{glob}} \leftarrow$ Eq. 2 for all queries ▷ block-centre keys, coarse positions
if $\sigma > \sigma_g$ **then return** O^{glob} ▷ opening phase
$O^{\text{loc}} \leftarrow$ Eq. 3 for image queries ▷ one FlexAttention call
return $[\, O^{\text{glob}}_{\mathcal{T}},\ \pi_h(\sigma)\, O^{\text{loc}} + (1 - \pi_h(\sigma))\, O^{\text{glob}}_{\mathcal{G}} \,]$ ▷ interpolated table; fp32 mix

Position ids in bf16. The FLUX.1 pipeline builds image position ids in bf16, which cannot represent every integer above 256, so positions beyond 256 tokens per axis are rounded. This affects every method, including full attention and AReA, at 2048×6144 (384 tokens per row). SCA takes the grid size from the sampler instead of inferring it from the rounded ids, so its window and subsampling geometry match the true grid.

Non-divisible grids. When H/n_0 is not an integer, the subsampling factor per axis is the divisor of H nearest to H/n_0, and the tile side the divisor nearest to t; neither case arises at the sizes we evaluate.

Table 6: **Cost.** Top: analytic attention FLOPs per image (57 blocks, 24 heads, head dimension 128, 28 steps, $T = 512$); "gated" skips the local branch in the first 8 steps; ratio is full over gated SCA. Bottom: measured time of one attention layer in bf16 on GH200, minimum over interleaved runs; SCA is both branches and the mix.

Size ($H \times W$)	Grid	(s_H, s_W)	Full	SCA, no gate	SCA, gated	Ratio
2048×4096	128×256	(2, 4)	2.17×10^{16}	5.97×10^{15}	5.12×10^{15}	4.2
4096×4096	256×256	(4, 4)	8.56×10^{16}	1.19×10^{16}	1.02×10^{16}	8.4
2048×6144	128×384	(2, 6)	4.84×10^{16}	8.93×10^{15}	7.66×10^{15}	6.3

Token grid	Pixels	Full (SDPA, ms)	SCA (ms)	Speedup
128×128	2048^2	7.1	5.2	1.4
256×256	4096^2	87.3	18.7	4.7
384×128	6144×2048	49.6	14.0	3.5

End-to-end latency. On one GH200, batch 1, 4096^2, 28 steps, after a warm-up generation, the medians over three images are (`results/latency_4096_batch1.csv`): full attention 184.6 s (184.1, 184.7, 184.6), AReA 212.6 s (207.3, 225.8, 212.6), SCA + band guard 88.1 s (88.1, 98.7, 88.1; single-pass compiled band), and SCA final 71.0 s (71.0, 83.1, 70.9). On the shared campaign nodes, per-image times were 71 s for SCA final, 120 s for SCA + band guard, and 262 s for AReA; the campaign code used an unfused band, so dropping it saved more there (48 s) than in the clean measurement (17 s). At the non-square sizes we have only campaign timings on shared nodes (mean over 191 prompts): 2048×4096, SCA 38.1 s and AReA 112.4 s; 2048×6144, SCA 55.8 s and AReA 150.5 s. The two methods ran on the same pool of nodes. [TODO: E3: clean latency at the other sizes, Table 3.]

D Hyperparameters

E Evaluation protocol and metric definitions

Prompts and references. The Aesthetic-4K evaluation split (Zhang et al., 2025a) has 195 prompts with 4K reference images. Four (boats, Moai, woman and bear, underwater surfer) were used to choose the opening phase and window and are excluded everywhere, leaving 191. The 46-prompt

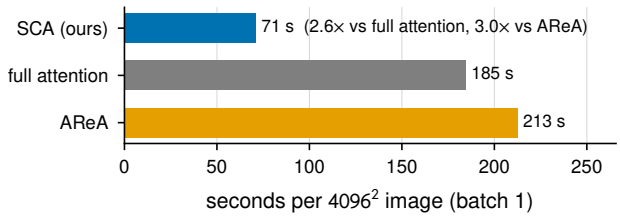

Figure 9: **SCA generates a 4096^2 image 2.6× faster than full attention and 3.0× faster than AReA.** Seconds per image at batch size 1 on an idle GH200, median over 3 prompts: SCA 71 s, full attention 185 s, and AReA (released code) 213 s.

Table 7: **All hyperparameters** of the main runs. Code commit `2d11740`; `scripts/sample.py -method sca -pi-table calib/k1_1024.npz -gate-sigma 0.89`.

Group	Setting	Value
Model	checkpoint, precision	FLUX.1-dev, bf16 weights and activations
	blocks, heads, head dim	19 double-stream + 38 single-stream, 24, 128
	native grid n_0, text tokens T	64 (1024^2 pixels, 16 pixels per token), 512
Sampler	steps, solver	28, flow-matching Euler
	guidance, shift	4.5 (distilled guidance), $\mu = 1.15$ at every size
	seed, decoding	seed 0 per prompt, shared across methods; tiled VAE decoding
Local branch	tile t, window w	8, 64 (clamped at borders), original positions
	kernel	FlexAttention; query blocks 8×16, key blocks 8×8, tile-major order
Global branch	subsampling	block centre, factor (s_H, s_W) = target grid / n_0
	positions	queries at $c(p)$, keys at block index, text at 0
Mixing	table	per (block, head, σ bin), $57 \times 24 \times 14$, scale-4 entry
	interpolation	linear in σ between bin centres, constant outside
	gate	$\sigma_g = 0.89$ (8 of 28 steps global only)
Calibration	prompts, resolution	32 at 1024^2 (16 more at 1280^2 for transfer)
	emulated scales	2 and 4, one pass; 14 bins of width $1/14$
	cost	about 1.5 GPU-hours on one GH200 in total
Band guard	(not in final method)	average-pool $p = 16$, bilinear upsampling

ablation subset is fixed before any ablation was run. HRGen-1K (Rajabi et al., 2026a) has 1000 prompts and no references; we use the first 200 [TODO: E4: confirm selection rule]. The 48 calibration prompts were written for this purpose and share no prompt with Aesthetic-4K (the largest word-set Jaccard overlap with any evaluation prompt is 0.26).

Preference and alignment metrics. ImageReward v1.0 (Xu et al., 2023) (bicubic resize to 224, centre crop). HPSv2 and HPSv2.1 (Wu et al., 2023): cosine of normalized image and text features with the official validation transform (longest side 224, padded). PickScore (Kirstain et al., 2023): $\exp(\text{logit scale})$ times the cosine. CLIP score (Radford et al., 2021): $100 \max(\cos, 0)$ with ViT-L/14. Aesthetic: the LAION improved aesthetic predictor (sac+logos+ava1, linear MSE) on the L2-normalized CLIP ViT-L/14 image embedding (Schuhmann, 2022).

No-reference quality. MUSIQ (Ke et al., 2021) (pyiqa defaults) on the full image at native resolution, and MUSIQ_c, its mean over four fixed random 1024^2 crops per image; crop positions depend only on the prompt id, so they are identical across methods. CLIP-IQA (Wang et al., 2022) and ARNIQA (Agnolucci et al., 2024) (pyiqa defaults) on the image downscaled with antialiased bicubic resampling to a long side of 1024 pixels, the scale both were trained on.

Distribution metrics. FID (Heusel et al., 2017) and KID between the 191 generated images and the 195 references, with Inception-v3 features (torch-fidelity) after a bilinear resize to 299^2. Patch

FID and KID use the same four 1024^2 crops per image on both sides (764 and 780 crops). Inception score uses the full image at 299^2, 10 splits.

Intervals. All intervals are 95% percentile bootstraps over prompts, paired: each draw resamples prompt ids and recomputes both methods' metric on the same ids, then their difference. Main comparisons use 1000 draws, ablations 2000 with seed 0 (`scripts/paired_bootstrap.py`). For set-level metrics (FID, KID, their patch versions, Inception score) the bootstrap resamples images within the evaluated set, which biases the absolute value upward because duplicates reduce diversity; the point estimate can therefore lie outside the interval of the absolute value, and for full attention's patch FID it lies just outside the interval of the difference too. We use these intervals only for the sign of differences.

F Extended results with intervals

Every table below is generated from the CSV files in `results/`; nothing is transcribed by hand.

Table 8: **All metrics at** 4096^2, 191 prompts, paired against AReA (released code). Δ is method minus AReA with its 95% paired bootstrap interval (1000 draws); **blue** / *orange*: significantly better / worse than AReA. Set-level metrics (FID, KID and their patch versions, Inception score) are computed on the whole set; their paired intervals resample prompts in both sets jointly. ‡Mean campaign wall-clock on shared nodes, not a clean latency measurement (Table 2 gives clean latency). Sources: `results/final4k_nb_*`, `final4k_*`, `final4k_full_*`.

Metric	AReA	SCA final	Δ [95% CI]	SCA + band	Δ [95% CI]	Full attention	Δ [95% CI]
ImageReward	0.917	1.229	+0.312 [+0.240, +0.384]	1.198	+0.281 [+0.208, +0.358]	−2.178	−3.095 [−3.216, −2.978]
Aesthetic	6.28	6.45	+0.17 [+0.12, +0.22]	6.46	+0.18 [+0.12, +0.24]	4.51	−1.77 [−1.87, −1.67]
CLIP score	27.67	29.11	+1.45 [+1.06, +1.82]	28.87	+1.20 [+0.79, +1.61]	15.38	−12.28 [−13.10, −11.44]
PickScore	22.35	23.09	+0.73 [+0.62, +0.86]	22.90	+0.54 [+0.40, +0.71]	17.94	−4.41 [−4.61, −4.18]
HPSv2 (×100)	28.64	29.77	+1.12 [+0.95, +1.31]	29.40	+0.75 [+0.53, +0.99]	20.75	−7.89 [−8.23, −7.54]
HPSv2.1 (×100)	27.20	29.71	+2.50 [+2.17, +2.88]	28.95	+1.75 [+1.31, +2.22]	7.61	−19.59 [−20.39, −18.71]
MUSIQ	39.4	45.3	+5.8 [+4.1, +7.5]	41.4	+2.0 [+0.0, +3.9]	22.9	−16.5 [−18.0, −15.0]
MUSIQ_c	50.4	61.2	+10.9 [+9.5, +12.4]	61.0	+10.6 [+9.2, +12.3]	33.2	−17.1 [−18.8, −15.4]
CLIP-IQA (×100)	67.5	72.9	+5.4 [+4.0, +6.8]	70.9	+3.3 [+1.8, +4.9]	49.0	−18.6 [−20.4, −16.7]
ARNIQA (×100)	67.3	71.2	+3.9 [+3.0, +4.9]	70.8	+3.5 [+2.4, +4.6]	54.5	−12.8 [−14.3, −11.2]
Inception score	6.42	6.52	+0.09 [−0.47, +0.49]	6.10	−0.33 [−0.88, +0.18]	3.83	−2.60 [−3.14, −1.83]
FID↓	146.0	138.4	−7.6 [−12.5, −3.5]	142.3	−3.7 [−9.6, +1.0]	312.7	+166.7 [+145.2, +170.9]
KID↓ ($\times 10^3$)	7.11	6.09	−1.02 [−2.93, +1.09]	5.48	−1.63 [−3.84, +0.51]	163.97	+156.86 [+139.17, +174.94]
FID_p ↓	110.2	106.5	−3.7 [−7.2, +1.1]	104.8	−5.4 [−9.1, −0.5]	257.9	+147.7 [+120.9, +144.4]
KID_p ↓ ($\times 10^3$)	11.60	9.24	−2.36 [−4.52, −0.12]	9.64	−1.96 [−4.39, +0.56]	172.65	+161.05 [+147.28, +174.21]
Seconds per image‡ ↓	261.7	71.5	−190.2 [−203.0, −176.8]	120.0	−141.7 [−155.8, −127.1]	196.5	−65.2 [−79.2, −50.5]

Table 9: **All metrics at the non-square sizes**, 191 prompts each, SCA final paired against AReA (released code). Conventions as in Table 8. Sources: `results/final_2048x4096_*`, `final_2048x6144_*`.

	2048 × 4096, factors (2, 4)			2048 × 6144, factors (2, 6)		
Metric	AReA	SCA	Δ [95% CI]	AReA	SCA	Δ [95% CI]
ImageReward	1.133	1.130	−0.003 [−0.079, +0.073]	0.389	0.664	+0.275 [+0.150, +0.400]
Aesthetic	6.41	6.21	−0.21 [−0.26, −0.15]	6.15	6.00	−0.15 [−0.25, −0.04]
CLIP score	28.51	28.21	−0.30 [−0.70, +0.11]	26.49	27.01	+0.52 [−0.01, +1.07]
PickScore	22.67	22.64	−0.02 [−0.13, +0.09]	21.51	21.74	+0.23 [+0.05, +0.43]
HPSv2 (×100)	28.38	28.55	+0.17 [+0.06, +0.28]	26.41	26.54	+0.13 [+0.00, +0.27]
HPSv2.1 (×100)	27.66	27.46	−0.20 [−0.49, +0.05]	23.35	23.48	+0.13 [−0.27, +0.54]
MUSIQ	51.8	55.9	+4.1 [+2.7, +5.7]	42.9	48.4	+5.5 [+3.6, +7.3]
MUSIQ_c	61.6	65.9	+4.3 [+3.3, +5.5]	53.1	60.6	+7.5 [+6.2, +9.0]
CLIP-IQA (×100)	74.8	73.5	−1.3 [−2.3, −0.4]	67.3	69.0	+1.6 [+0.2, +3.1]
ARNIQA (×100)	73.0	72.8	−0.2 [−0.9, +0.5]	69.3	70.2	+0.9 [+0.1, +1.7]
Inception score	6.21	6.65	+0.44 [−0.16, +0.95]	5.36	6.35	+0.99 [+0.13, +1.22]
FID↓	144.8	143.8	−1.1 [−5.9, +4.3]	163.6	143.8	−19.8 [−24.5, −11.9]
KID↓ ($\times 10^3$)	7.95	7.87	−0.09 [−2.86, +2.90]	18.08	8.46	−9.62 [−13.59, −5.98]
FID_p ↓	119.0	124.1	+5.1 [+1.2, +9.1]	112.8	116.5	+3.7 [−0.2, +11.0]
KID_p ↓ ($\times 10^3$)	14.51	16.31	+1.79 [+0.34, +3.46]	13.85	15.18	+1.33 [−2.28, +5.16]
Seconds per image‡ ↓	112.4	38.1	−74.3 [−76.6, −72.0]	150.5	55.8	−94.7 [−96.9, −92.4]

Table 10: **Ablation intervals for Table 5**, 46 prompts at 4096^2, paired against SCA final (2000 bootstrap draws). The first column gives SCA final's absolute value; the last row gives patch FID per variant (set-level, no paired interval). Sources: `results/ablnb_*`.

Metric	SCA final	$\pi = 1$	shared $\pi(\sigma)$	per-layer	$\pi = 0$	mean pool	v0, no gate
ImageReward	1.124	-0.357 [-0.524, -0.207]	-0.090 [-0.173, -0.012]	-0.054 [-0.159, $+0.030$]	-0.395 [-0.507, -0.295]	-1.518 [-1.847, -1.193]	-1.328 [-1.595, -1.064]
Aesthetic	6.45	-0.21 [-0.35, -0.07]	-0.09 [-0.18, -0.02]	$+0.01$ [-0.06, $+0.07$]	-0.76 [-0.87, -0.64]	-0.61 [-0.77, -0.44]	-0.08 [-0.17, $+0.01$]
CLIP score	29.41	-1.93 [-2.68, -1.20]	-0.38 [-0.78, $+0.08$]	-0.13 [-0.54, $+0.28$]	-3.48 [-4.43, -2.59]	-3.59 [-4.76, -2.49]	-1.65 [-2.58, -0.84]
PickScore	22.96	-1.20 [-1.44, -0.99]	-0.38 [-0.51, -0.26]	-0.16 [-0.28, -0.05]	-1.44 [-1.66, -1.22]	-2.60 [-3.00, -2.17]	-2.18 [-2.51, -1.85]
HPSv2 ($\times 100$)	29.71	-1.78 [-2.07, -1.50]	-0.96 [-1.19, -0.73]	-0.50 [-0.63, -0.37]	-3.32 [-3.72, -2.94]	-4.04 [-4.66, -3.43]	-1.89 [-2.34, -1.45]
HPSv2.1 ($\times 100$)	29.98	-2.86 [-3.35, -2.36]	-1.83 [-2.24, -1.43]	-0.70 [-0.98, -0.41]	-5.66 [-6.35, -4.98]	-9.75 [-11.30, -8.17]	-4.07 [-5.11, -3.04]
MUSIQ	46.1	-3.5 [-6.5, -0.7]	-4.9 [-7.9, -1.9]	$+0.7$ [-1.6, $+3.0$]	-3.5 [-6.7, -0.2]	-1.8 [-5.3, $+1.9$]	-5.6 [-9.0, -2.1]
MUSIQ$_c$	62.5	-0.3 [-2.5, $+1.7$]	-10.4 [-12.8, -8.3]	-2.4 [-3.6, -1.1]	-11.7 [-13.6, -9.8]	-4.2 [-8.4, -0.5]	$+2.5$ [-0.4, $+5.5$]
CLIP-IQA ($\times 100$)	71.3	-4.6 [-6.3, -3.0]	-7.7 [-9.4, -6.0]	-0.9 [-2.0, $+0.1$]	-27.0 [-30.2, -23.8]	-10.1 [-13.0, -7.1]	$+2.9$ [$+1.0$, $+4.9$]
ARNIQA ($\times 100$)	70.5	-1.0 [-2.0, $+0.0$]	-6.5 [-8.5, -4.8]	-1.5 [-2.3, -0.9]	-45.1 [-47.6, -42.5]	-4.8 [-6.7, -2.6]	$+4.3$ [$+2.7$, $+6.1$]
FID$_p$ ↓	170.2	194.3	169.1	170.2	238.9	187.5	204.8

Table 11: **Intervals for the band-guard ablations** (Table 14 and further variants), 46 prompts at 4096^2, paired against SCA + band guard, whose absolute values form the first column. Here patch FID has paired intervals. Sources: `results/abl_*`, `route_*`.

Metric	Reference	no band	no gate	shared $\pi(\sigma)$	per-layer	$\pi = 1$
ImageReward	1.165	-0.041 [-0.109, $+0.021$]	-0.357 [-0.511, -0.222]	-0.032 [-0.084, $+0.015$]	-0.024 [-0.055, $+0.004$]	-0.015 [-0.075, $+0.040$]
Aesthetic	6.43	$+0.01$ [-0.05, $+0.08$]	-0.23 [-0.35, -0.11]	-0.06 [-0.13, $+0.00$]	$+0.03$ [-0.01, $+0.06$]	$+0.03$ [-0.02, $+0.10$]
CLIP score	29.01	$+0.40$ [-0.04, $+0.92$]	-1.15 [-1.82, -0.47]	-0.09 [-0.53, $+0.31$]	-0.03 [-0.28, $+0.23$]	$+0.16$ [-0.19, $+0.53$]
PickScore	22.71	$+0.25$ [$+0.09$, $+0.41$]	-0.86 [-1.13, -0.59]	$+0.02$ [-0.10, $+0.14$]	$+0.01$ [-0.06, $+0.08$]	$+0.12$ [$+0.03$, $+0.22$]
HPSv2 ($\times 100$)	29.18	$+0.54$ [$+0.31$, $+0.75$]	-0.49 [-0.87, -0.14]	-0.20 [-0.42, $+0.01$]	-0.08 [-0.19, $+0.04$]	$+0.16$ [$+0.04$, $+0.29$]
HPSv2.1 ($\times 100$)	28.81	$+1.17$ [$+0.71$, $+1.64$]	-1.39 [-2.18, -0.63]	-0.33 [-0.68, $+0.01$]	-0.01 [-0.19, $+0.19$]	$+0.55$ [$+0.30$, $+0.79$]
MUSIQ	43.1	$+2.9$ [$+0.7$, $+5.7$]	-0.7 [-3.6, $+2.4$]	$+0.5$ [-2.5, $+4.0$]	-0.3 [-1.9, $+1.8$]	$+1.0$ [-0.3, $+2.6$]
MUSIQ$_c$	62.1	$+0.4$ [-0.9, $+1.6$]	-1.0 [-3.4, $+1.7$]	-5.7 [-7.2, -4.5]	-1.5 [-2.4, -0.7]	$+0.9$ [$+0.3$, $+1.4$]
CLIP-IQA ($\times 100$)	69.6	$+1.7$ [$+0.3$, $+3.2$]	-0.6 [-2.7, $+1.5$]	-6.5 [-8.4, -4.7]	-1.2 [-2.1, -0.5]	$+0.5$ [-0.9, $+1.9$]
ARNIQA ($\times 100$)	69.6	$+0.9$ [-0.6, $+3.1$]	$+1.6$ [-0.4, $+4.5$]	-4.6 [-7.1, -2.9]	-1.2 [-2.0, -0.6]	$+0.6$ [-0.1, $+1.4$]
FID$_p$ ↓	169.3	$+0.8$ [-4.4, $+6.4$]	$+9.0$ [$+1.3$, $+17.9$]	-4.8 [-10.7, -0.5]	-1.4 [-4.6, $+1.8$]	$+1.3$ [-3.3, $+5.8$]

Pending comparisons. [TODO: E1: per-metric intervals for DyPE, SEGA, UltraImage vs. AReA and vs. SCA.] [TODO: E2: 6144^2 intervals.] [TODO: E4, E5, E6, E7, E8, E9: per-metric intervals for every row of Table 4.]

G Calibration table at non-square sizes

At $(s_H, s_W) = (2, 4)$ the two axes have different factors, and SCA uses the table of the emulated scale nearest the larger factor, scale 4. On the 46 ablation prompts at 2048×4096, the scale-2 table changes the paired metrics negligibly (Table 13): HPSv2 $+0.09$, HPSv2.1 $+0.17$ (both $\times 100$), and PickScore $+0.05$ are significant but small, and ImageReward ($+0.01$ $[-0.03, 0.05]$) and the other metrics are within noise. The table choice therefore does not explain why SCA only ties AReA at this size, and we keep the default rule.

H The low-band duplication guard

Construction. Let LP_p average-pool a head's image output over $p \times p$ tokens and upsample it bilinearly, and $\mathrm{HP}_p = \mathrm{I} - \mathrm{LP}_p$. The guard replaces the low band of the mix M by that of the global branch:

$$O \;=\; \mathrm{LP}_p\big(O^{\mathrm{glob}}\big) + \mathrm{HP}_p\big(M\big) \;=\; O^{\mathrm{glob}} + \pi_h(\sigma)\,\mathrm{HP}_p\big(O^{\mathrm{loc}} - O^{\mathrm{glob}}\big). \tag{6}$$

The local branch then contributes only its difference from the global branch above the cutoff. The cutoff should lie below $1/(2s)$, where the global branch aliases, and above $1/w$, below which the window cannot resolve; with $w = 64$ and $s = 4$ this leaves periods of 8 to 64 tokens. We used $p = 16$; $p = 8$ and $p = 32$ were worse on four development prompts, and a noise-dependent cutoff behaved like $p = 16$. FreeScale (Qiu et al., 2024) also fuses two self-attention outputs by frequency band, in the opposite direction: it keeps the blurred low band of attention over overlapping crops and takes the high band from full self-attention at the target size. The guard instead takes the low band from the global branch, which stays at native token count and offsets, and keeps the per-head weight on the high band.

History: what it fixes and what it costs. On the development prompts, the band alone (no opening phase) fixed the boats but left extra statues and faces on the Moai and woman-and-bear prompts; the opening phase alone got the main subject right, but small extra figures appeared late on some

Table 12: **Band-guard ablations, continued.** Conventions as in Table 11.

Metric	Reference	route 0.5	route 0.3	mean pool	$\pi = 0$	v0
ImageReward	1.165	−0.051 [−0.095, −0.009]	−0.026 [−0.082, +0.024]	−0.848 [−1.083, −0.615]	−0.436 [−0.537, −0.336]	−1.369 [−1.633, −1.113]
Aesthetic	6.43	−0.02 [−0.08, +0.04]	+0.02 [−0.03, +0.07]	−0.39 [−0.54, −0.25]	−0.75 [−0.86, −0.63]	−0.07 [−0.15, +0.03]
CLIP score	29.01	−0.00 [−0.31, +0.28]	+0.13 [−0.23, +0.51]	−2.96 [−4.18, −1.67]	−3.08 [−3.98, −2.16]	−1.25 [−2.02, −0.40]
PickScore	22.71	+0.04 [−0.03, +0.12]	+0.09 [+0.01, +0.17]	−1.80 [−2.10, −1.49]	−1.19 [−1.41, −0.96]	−1.94 [−2.23, −1.65]
HPSv2 (×100)	29.18	+0.02 [−0.11, +0.15]	+0.11 [−0.01, +0.24]	−2.35 [−2.81, −1.94]	−2.79 [−3.18, −2.45]	−1.35 [−1.74, −0.95]
HPSv2.1 (×100)	28.81	+0.18 [−0.04, +0.40]	+0.34 [+0.12, +0.58]	−5.81 [−6.96, −4.74]	−4.48 [−5.17, −3.80]	−2.90 [−3.82, −2.01]
MUSIQ	43.1	+1.2 [−0.2, +3.0]	+0.8 [−0.4, +2.5]	−1.8 [−5.5, +1.7]	−0.6 [−4.4, +3.4]	−2.7 [−6.1, +0.7]
MUSIQ_c	62.1	+0.5 [−0.0, +1.1]	+0.8 [+0.2, +1.4]	−0.8 [−5.0, +3.6]	−11.3 [−13.3, −9.6]	+2.9 [−0.3, +6.2]
CLIP-IQA (×100)	69.6	+0.3 [−1.3, +1.5]	+0.6 [−0.7, +1.7]	−5.4 [−9.3, −1.6]	−25.2 [−29.1, −21.4]	+4.6 [+2.1, +7.3]
ARNIQA (×100)	69.6	+0.1 [−0.6, +0.8]	+0.6 [−0.1, +1.3]	−1.1 [−3.6, +1.9]	−44.3 [−47.4, −40.7]	+5.1 [+2.8, +8.1]
$\text{FID}_p \downarrow$	169.3	+1.5 [−2.5, +5.0]	+1.1 [−3.6, +5.8]	+18.8 [+11.6, +28.3]	+69.5 [+46.8, +70.9]	+35.5 [+25.8, +44.7]

Table 13: **Calibration table at** 2048×4096, 46 prompts, paired against SCA final with the scale-4 table. Sources: `results/ps2_*`.

Metric	SCA, scale-2 table	AReA (released code)
ImageReward	+0.012 [−0.025, +0.050]	+0.027 [−0.141, +0.192]
Aesthetic	−0.00 [−0.03, +0.02]	+0.29 [+0.16, +0.43]
CLIP score	−0.02 [−0.22, +0.17]	+0.19 [−0.53, +0.91]
PickScore	+0.05 [+0.00, +0.09]	+0.14 [−0.09, +0.39]
HPSv2 (×100)	+0.09 [+0.04, +0.13]	−0.19 [−0.44, +0.06]
HPSv2.1 (×100)	+0.17 [+0.08, +0.26]	+0.39 [−0.21, +1.01]
MUSIQ	+0.0 [−0.5, +0.5]	−3.1 [−7.5, +1.2]
MUSIQ_c	−0.0 [−0.4, +0.2]	−4.1 [−6.7, −1.3]
CLIP-IQA (×100)	−0.6 [−1.2, +0.0]	+1.3 [−0.8, +3.4]
ARNIQA (×100)	−0.2 [−0.6, +0.1]	+0.4 [−1.5, +2.8]

prompts; both together gave one subject on the boats, Moai, and woman-and-bear prompts. This was our first full configuration. On 46 prompts, however, removing the guard improves HPSv2 (+0.54), HPSv2.1 (+1.17), PickScore (+0.25), CLIP-IQA (+1.7), and MUSIQ (+2.9), all significant, leaves the other metrics within noise, and saves 17 s per image on an idle GPU. The guard also hid what the per-head mix contributes. With it (Table 14), local only matched the per-head mix within 2% on preference metrics, because the guard replaces the low band of every head's output with that of the global branch whatever the head's weight, so it injects global low frequencies even at $\pi = 1$; without the guard local only loses on nearly every metric (Section 4.2). Likewise, with the guard one shared weight and per-layer weights matched the per-head table on preference metrics and lost only on no-reference quality, and hard per-head routing (10 to 17% of heads global at threshold 0.5, about 6% at 0.3) stayed within 2% of the mix. The guard is an option for prompts where late duplication is visible, not part of the default method.

I Reproducing AReA

We run AReA's released code with its FLUX.1 defaults (block-centre pooling, first phase for 8 of 28 steps, halo width 8, bridge ratio 0.5), the FLUX.1-dev checkpoint, guidance 4.5, 28 steps, μ at its maximum, and the seeds shared with the other methods. AReA's paper reports ImageReward 1.32 at 4K under its own settings; our shared protocol gives 0.92, so cross-paper numbers are not comparable, while all numbers in this paper come from one protocol. AReA's paper reports 151 s for AReA and 314 s for full attention at 4096^2 on an H100; we measure 212.6 s and 184.6 s on one GH200 (medians of three images). We use commit `9e38ec2` of the released repository, which includes a cost cap on halo and bridge tokens and a jagged halo edge; neither appears in the paper text, and the paper does not say whether its Table 1 used them. Recomputed from the code with 512 text tokens, the first-phase pooling factors are 4, 4, 4, 4, 2, 2, 2, 2 at 4096×4096 and 2048×4096, and 8, 8, 8, 4, 4, 4, 2, 2 at 2048×6144. The second-phase windows shrink over 20 steps from 180, 151, and 192 tokens respectively to 64. At 2048×6144 no window is cheaper than full attention once the text tokens are counted, so the first window falls back to 192 tokens and the cap drops the bridge and halo on the first three second-phase steps. Other differences from the release defaults are the FLUX.1-dev checkpoint instead of FLUX.1-Krea-dev, and one seed per prompt shared across methods instead of one seed for all prompts. [TODO: E9: AReA with its default FLUX.1-Krea-dev checkpoint, against recalibrated SCA.]

Table 14: **Ablations around the band-guard configuration at** 4096^2, 46 prompts. The reference row gives absolute values for SCA + band guard (band 16); every other row changes one thing, keeps the guard unless stated, and reports its paired difference to the reference. Marks and units as in Table 5; here patch FID also has paired intervals.

Variant	Mix err.	IR	Aesth.	HPSv2	HPSv2.1	PickS.	CLIP	CLIP-IQA	ARNIQA	MUSIQ	MUSIQ_e	$\text{FID}_p \downarrow$
SCA + band guard (reference)	0.200	1.17	6.43	29.18	28.81	22.71	29.01	69.6	69.6	43.1	62.1	169.3
no band (**= SCA final**)	0.200	−0.04	+0.01	+0.54	+1.17	+0.25	+0.40	+1.7	+0.9	+2.9	+0.4	+0.8
no gate	0.200	−0.36	−0.23	−0.49	−1.39	−0.86	−1.15	−0.6	+1.6	−0.7	−1.0	+9.0
one shared $\pi(\sigma)$	0.230	−0.03	−0.06	−0.20	−0.33	+0.02	−0.09	−6.5	−4.6	+0.5	−5.7	−4.8
per-layer $\pi_\ell(\sigma)$	0.223	−0.02	+0.03	−0.08	−0.01	+0.01	−0.03	−1.2	−1.2	−0.3	−1.5	−1.4
local only ($\pi = 1$)	0.261	−0.01	+0.03	+0.16	+0.55	+0.12	+0.16	+0.5	+0.6	+1.0	+0.9	+1.3
hard routing, threshold 0.5	n/a	−0.05	−0.02	+0.02	+0.18	+0.04	0.00	+0.3	+0.1	+1.2	+0.5	+1.5
hard routing, threshold 0.3	n/a	−0.03	+0.02	+0.11	+0.34	+0.09	+0.13	+0.6	+0.6	+0.8	+0.8	+1.1
mean pooling	n/a	−0.85	−0.39	−2.35	−5.81	−1.80	−2.96	−5.4	−1.1	−1.8	−0.8	+18.8

J Qualitative galleries

Galleries are uncurated unless stated: prompts are drawn at random from the evaluation prompts, excluding the development prompts and those shown in the main text.

[TODO: E1: gallery of DyPE, SEGA, UltraImage, AReA, SCA at 4096^2.] [TODO: E2: 6144^2 gallery.] [TODO: E2: 8192^2 gallery with 1:1 crops.] [TODO: E8: FLUX.2-dev gallery.]

K Failure cases and limitations

Late duplication. With the final method, small extra figures can appear on some prompts in the last 20 steps, when the local branch dominates most heads. The band guard suppresses some of them but costs preference scores.

Articulated subjects. The underwater-surfer development prompt keeps doubled limbs in every variant; AReA's image shows a large translucent ghost.

Small factors. At 2048×4096 the window covers half the image height, so the local branch alone already sees most of the image along that axis, and the per-head mix gains little over AReA's shrinking windows; aesthetic score and patch FID are worse.

Text-image offsets. The local branch keeps out-of-range text-image offsets (Remark 1).

Metrics. No-reference quality metrics reward sharp repeated subjects (v0 scores higher on CLIP-IQA and ARNIQA), and FID against 195 references at 299^2 resolution sees little of the 4K detail; we therefore report every metric and the qualitative galleries.

L Compute and reproducibility

Compute. All runs used single NVIDIA GH200 GPUs (96 GB) on a shared cluster. Calibration took about 1.5 GPU-hours. Generation at 4096^2 takes about 71 s per image for SCA, 262 s for AReA, and 197 s for full attention on shared nodes (campaign means); the main comparisons, ablations, and non-square sizes comprise more than 2,000 generated images. [TODO: Lead: total GPU-hours including E1 to E9, from the scheduler accounting.]

Reproducibility. The code (attention processor, calibration, sampling, evaluation wrappers, AReA wrapper, and tests), the calibration table `k1_1024.npz`, the per-image metric files, and the per-comparison summary and paired-delta CSV files behind every table will be released. All main-text numbers are generated from those CSV files. Spool tasks ran from frozen copies of the code at a fixed commit, and each image's sidecar records the settings and a code fingerprint.

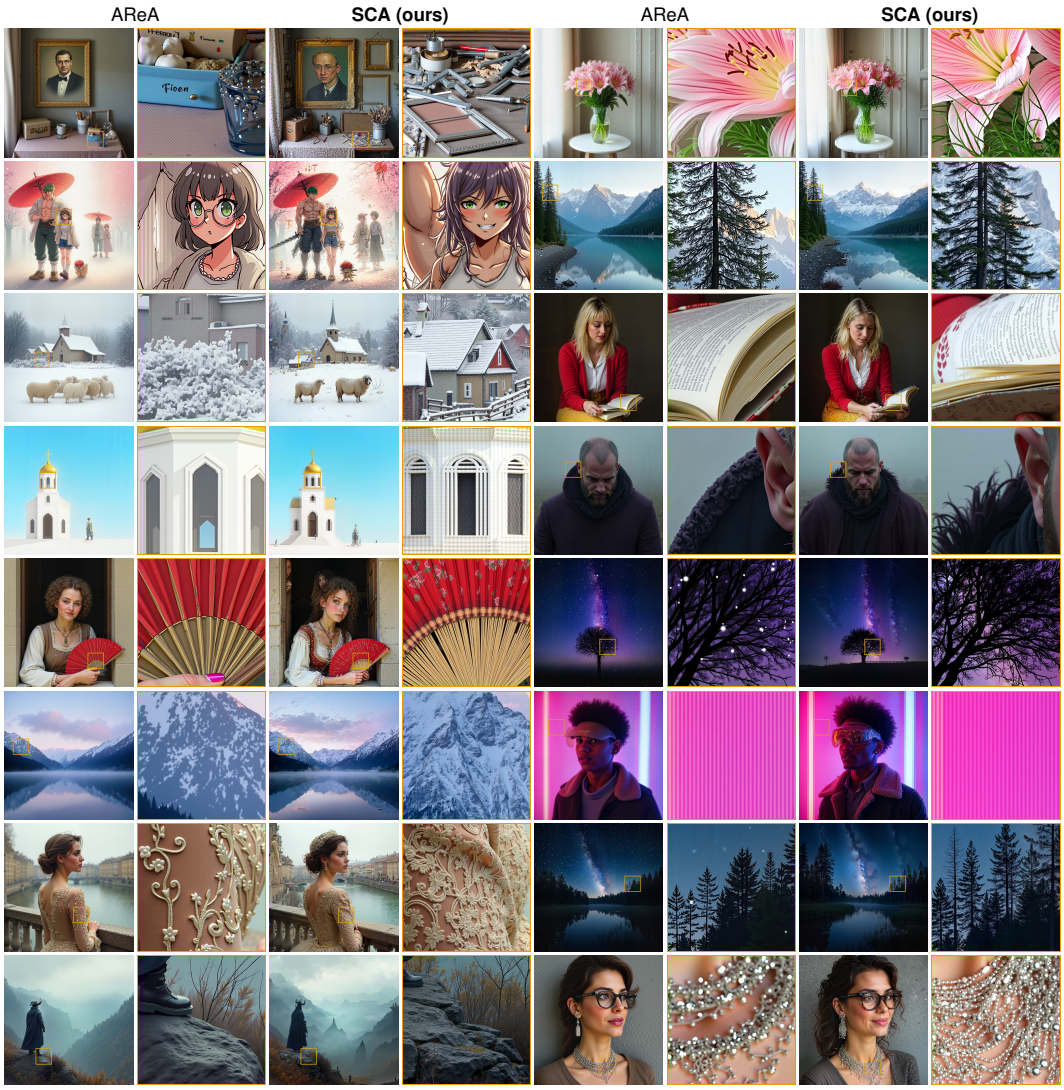

Figure 10: **Random prompts at 4096×4096, AReA vs SCA (ours).** 32 prompts sampled uniformly at random (seed 0) from the evaluation prompts, excluding the development prompts and those shown in the main text; not curated. Each full image is followed by a 1:1 crop of its outlined window, chosen automatically as the region with the most edge energy averaged over the methods. FLUX.1-dev, seed 0.

M BROADER IMPACT

SCA lowers the cost of generating very large images with existing open-weight models, which helps applications such as print, design, and scientific visualization, and lowers the energy per image relative to full attention. The same reduction lowers the cost of producing high-resolution synthetic images that could be used to mislead. SCA adds no capability beyond the base model's and does not change its safety filters or watermarking; the risks and mitigations of the base model apply unchanged.

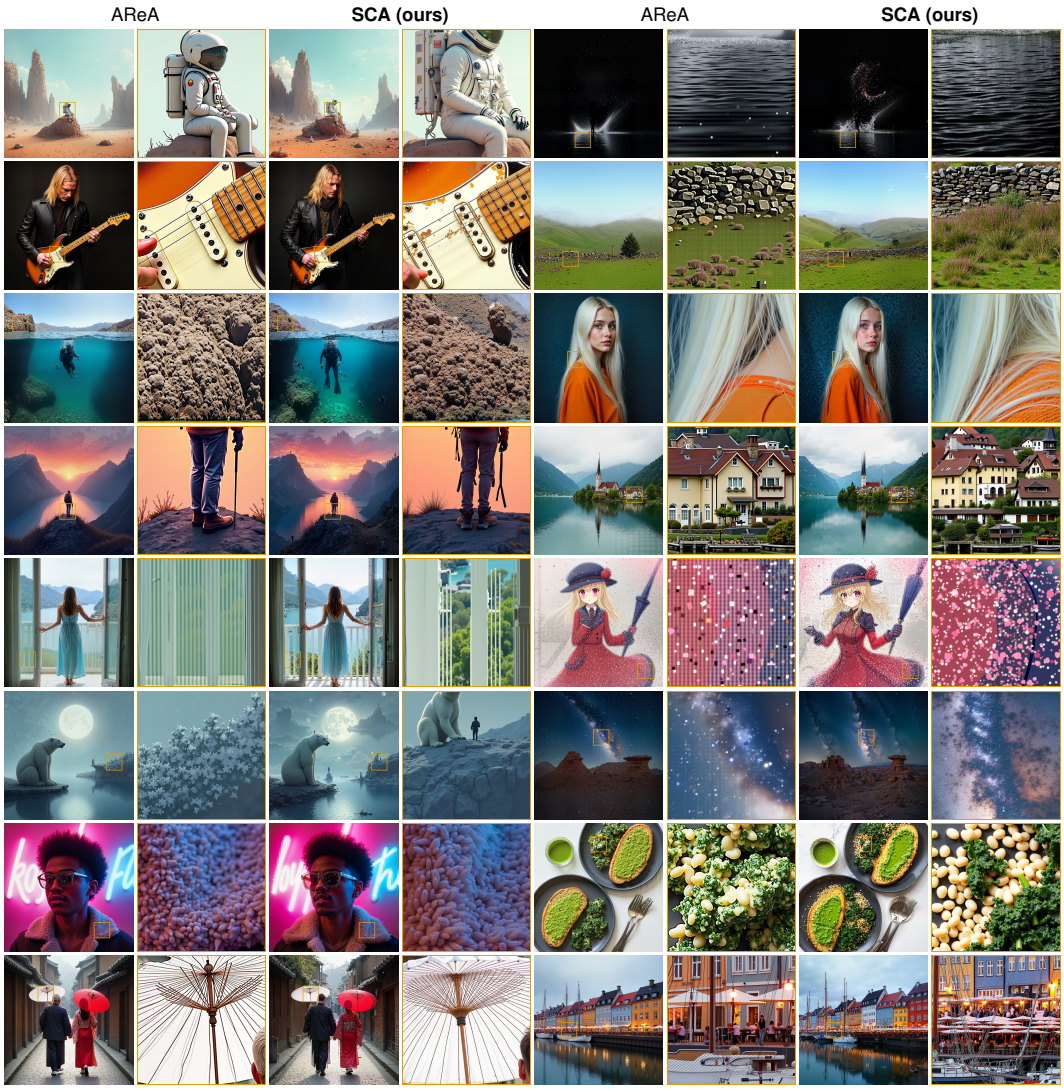

Figure 11: **Random prompts at 4096×4096, AReA vs SCA (ours) (continued).** Same protocol as Figure 10.

Figure 12: **Random prompts at 4096×2048 (width×height), AReA vs SCA (ours).** 6 prompts sampled at random from the evaluation prompts, disjoint from the other galleries and excluding the development prompts and those in the main text; not curated. Each image is followed by a 1:1 crop of its outlined 512×512 window, chosen automatically as the region with the most edge energy averaged over the methods. FLUX.1-dev, seed 0.

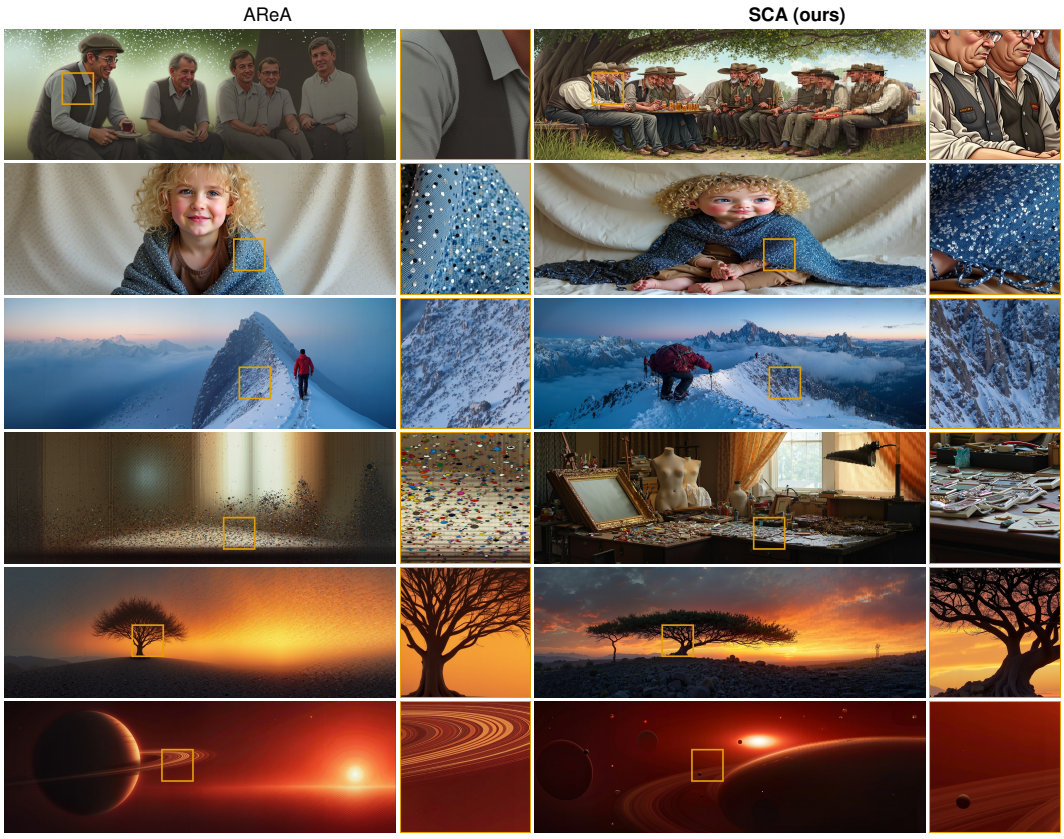

Figure 13: **Random prompts at 6144×2048 (width×height), AReA vs SCA (ours).** 6 prompts sampled at random from the evaluation prompts, disjoint from the other galleries and excluding the development prompts and those in the main text; not curated. Each image is followed by a 1:1 crop of its outlined 512×512 window, chosen automatically as the region with the most edge energy averaged over the methods. FLUX.1-dev, seed 0.

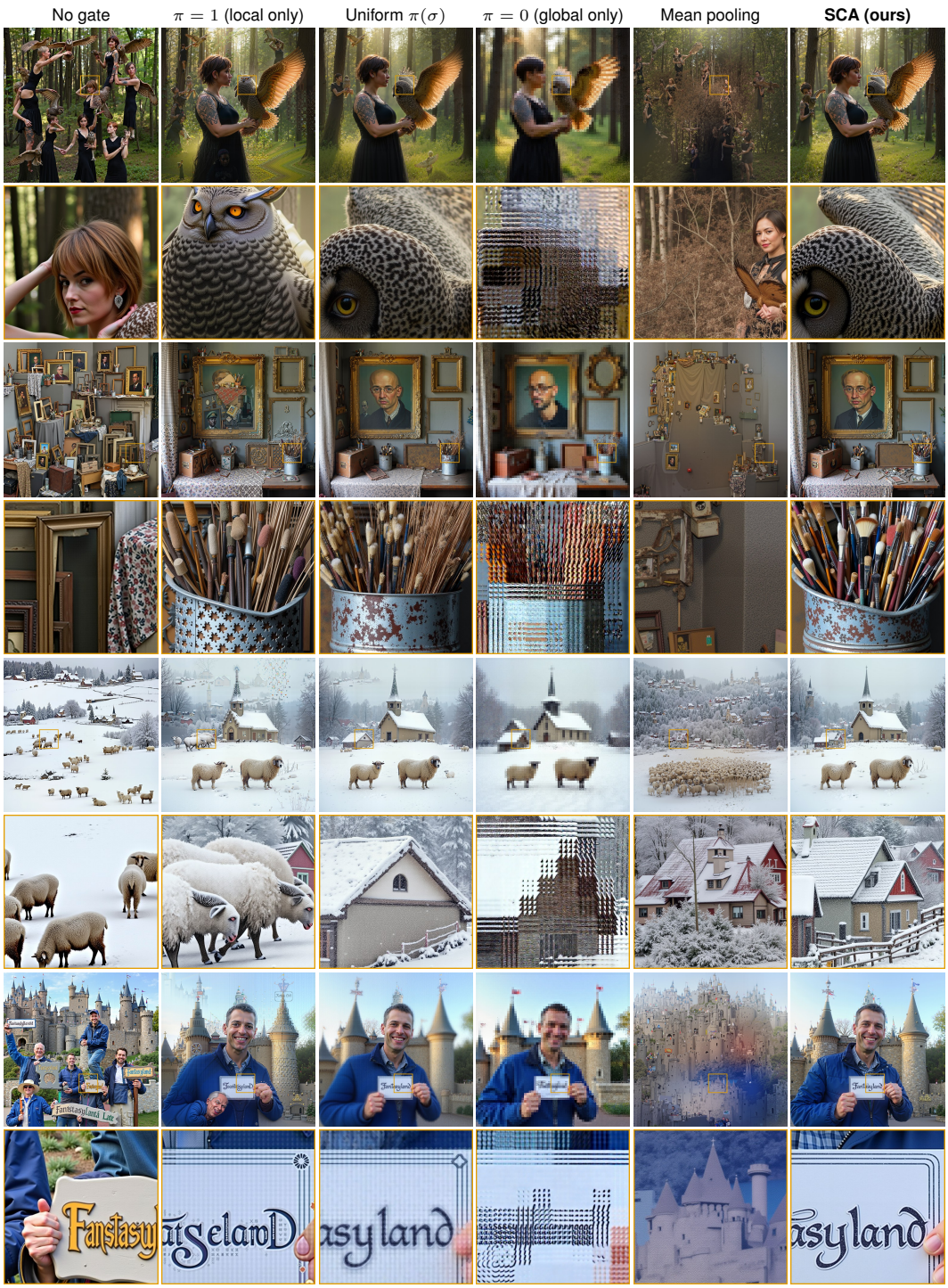

Figure 14: **Removing the gate duplicates subjects; dropping either branch or the per-head weights costs detail or coherence.** Four ablation prompts at 4096×4096, seed 0; each prompt shows the full images and, below them, 1:1 crops of the outlined 512×512 window (chosen automatically as the region with the most edge energy averaged over the six variants). No gate: the calibrated mix from the first step. $\pi = 1$ and $\pi = 0$: after the gate, the local or the global branch alone. Uniform $\pi(\sigma)$: one weight shared by all heads. Mean pooling: average-pooled instead of subsampled global keys. Without the gate the subject repeats across the canvas. $\pi = 0$ crops show the coarse grid of the pooled branch, and mean pooling loses the layout. $\pi = 1$ and uniform $\pi(\sigma)$ look close to SCA in these crops; their deficits show in the metrics.

Figure 15: **Failure modes.** Top: without the gate, local windows decide the layout at high noise and duplicate the subject (no gate, 64-token window; no gate, 16-token window, which gives smaller and more numerous copies); SCA keeps one subject. Two development prompts, 4096×4096. Middle and bottom: errors SCA still makes on evaluation prompts, with 1:1 crops of the outlined window and the same window from AReA. A thin diagonal sword is drawn as a staircase of short blocks; a patch beside the coat turns into a coarse grid; a dancer's rear arm ends in a foot; and the border of a dense illustration dissolves into speckle. AReA's images show none of these four errors, though its compositions differ, so its crops are not of the same objects. All seed 0.